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Model-Based Identifiable Parameter Determination Applied to a Simultaneous Saccharification and Fermentation Process Model for Bio-Ethanol Production

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In this work, a methodology for the model-based identifiable parameter determination (MBIPD) is presented. This systematic approach is proposed to be used for structure and parameter identification of nonlinear models of biological reaction networks. Usually, this kind of problems are over-parameterized with large correlations between parameters. Hence, the related inverse problems for parameter determination and analysis are mathematically ill-posed and numerically difficult to solve. The proposed MBIPD methodology comprises several tasks: (i) model selection, (ii) tracking of an adequate initial guess, and (iii) an iterative parameter estimation step which includes an identifiable parameter subset selection (SsS) algorithm and accuracy analysis of the estimated parameters. The SsS algorithm is based on the analysis of the sensitivity matrix by rank revealing factorization methods. Using this, a reduction of the parameter search space to a reasonable subset, which can be reliably and efficiently estimated from available measurements, is achieved. The simultaneous saccharification and fermentation (SSF) process for bio-ethanol production from cellulosic material is used as case study for testing the methodology. The successful application of MBIPD to the SSF process demonstrates a relatively large reduction in the identified parameter space. It is shown by a cross-validation that using the identified parameters (even though the reduction of the search space), the model is still able to predict the experimental data properly. Moreover, it is shown that the model is easily and efficiently adapted to new process conditions by solving reduced and well conditioned problems. © 2013 American Institute of Chemical Engineers *Biotechnol. Prog.*, 29:1064–1082, 2013

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Introduction

The determination of parameter values as well as the validation of mechanistic chemical/biochemical models involves the numeric solution of parameter estimation (PE) problem. This task is often difficult for several reasons[1–3]:

a. The process models frequently are over-parameterized;

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b. The process models are highly nonlinear and the optimization algorithms can exhibit convergence problems;
c. The available experimental data set is often limited (i.e., the number of experimental runs or measurements is small, the experimental design is highly correlated, or data are noisy);
d. Effects of some parameters on model predictions may be small, or the effects of some parameters may be correlated with the effects of others;
e. Poor initial guess (IG) are used in the PE problem.
f. The lack of identifiability or difficulty of assigning unique values for the unknown parameters.

If some of the above problems exist, it is necessary to decide whether the model structure must be changed (to lump parameters, to remove/add terms from equations, etc.), new experiments must be conducted with a specific design of experiment, or a subset of parameters should be estimated while the others are fixed at some appropriate values (e.g., information from literature, earlier data or PE analysis, or knowledge of physically realistic parameter values).[1,2,4,5]

Identifiability and distinguishability of nonlinear parametric models are classical elements in model identification.[4,6,7] Especially interesting for practical applications is the detection of correlations between parameters as well as the identification of the identifiable parameter subset which guarantees well-posed PE problems.[8] Contributions can be found where eigenvalue decomposition (EVD) and singular value decomposition (SVD) methods are applied to the Jacobian or Hessian of nonlinear least squares PE problems.[4,9–12] Different authors focused on correlation and collinearity index methods,[13,14] orthogonalization,[15] clustering,[16] methods based on scalar measures of the Fisher information matrix (FIM)[17] and mean-squared-error methods[18,19] for the parameter subset selection (SsS). When a model is unidentifiable, there are two options: (1) a model simplification,[4,12] i.e., fixing nonidentifiable parameters to nominal values or simplifying model structure by focusing on parts including nonidentifiable parameters,[20] (2) design informative experiments,[2,21] i.e., using optimization to find experiments that maximize identifiability and thereby maximize parameter accuracy.

Due to the nonlinearity of bioprocess models, the PE problem most likely contains multiple local minima. However, applying gradient-based nonlinear least squares estimation algorithms, the finding of the global solution depends mainly on the selection of the initial parameter guess. Computed solutions are usually checked by rerunning the estimation routine with the obtained parameter set as new IG or starting from different initial values. However, correlation between model parameters constitutes an obstacle to determine a unique minimizing parameter set.[22]

The investigation of dynamics of biochemical pathways based on mathematical modeling and supported by quantitative experimental data has been increasingly adopted.[23–27] This approach is suitable for supporting the biologists in the process of hypothesis generation and the design of experiments, decreasing the experimental effort necessary to characterize a biological system.[5,28] Moreover, models for the behavior of microbiological systems can represent a valuable tool in the industrial context, as these can be used for process design (i.e., development of new process configurations), operation (i.e., exploring new optimized operating conditions), and for identifying opportunities of new technologies.

This contribution is aimed at the parameterization and validation of a highly nonlinear process model for the simultaneous saccharification and fermentation (SSF) process for producing ethanol from lignocellulosic waste materials. Bioethanol has been an excellent candidate for the replacement of traditional fossil fuels by alternative sources of liquid fuels. However, bio-ethanol made from sugar sources, such as starch, corn, or sugar cane, has not been competitive with traditional fuels from petrol because of relatively high costs and the complicated situation of the food security. Therefore, alternative raw materials, for instance (ligno) cellulosic waste materials such as crop residues, grass, wood ships, wheat straw, or bagasse, which can act as an abundant and cheap source of fermentable sugars, have become increasingly interesting.[16,23,29,30]

Experimental data were taken from the SSF process using sugarcane bagasse as biomass.[30] The SSF process combines enzymatic hydrolysis with ethanol fermentation to keep the concentration of glucose low. Among the various cellulose bioconversion schemes, this process is considered to be the most promising regarding its efficiency.[16,24] If fermentation occurs simultaneously with saccharification, glucose produced during the saccharification will be rapidly converted into ethanol, reducing the inhibition caused by high sugar concentration, and therefore, the hydrolysis rate can be accelerated.[31] In comparison with the process where these two stages are sequential, the SSF method enables attainment of higher (up to 40%) yields of ethanol by removing end-product inhibition, as well as by eliminating the need for separate reactors for saccharification and fermentation.[29] Other advantages of this SSF are an easier operation, a shorter fermentation time, a reduced risk of contamination with undesired microorganisms, due to the high temperature of the process, the presence of ethanol in the reaction medium, and the anaerobic conditions and a lower equipment requirement than the sequential process because no hydrolysis reactors are needed.[29,32] In spite of the clear advantages presented by the SSF, the inconvenience is that there exist different optimal conditions for hydrolysis and fermentation, which implies a difficult control and optimization of process parameters.[32] Besides, ethanol itself and some toxic substances arising from the pretreatment of the lignocelluloses inhibit the action of fermenting microorganisms, as well as the cellulase activity.[29] On the other hand, some compounds (e.g., proteolytic enzymes) that are released on cell lysis or are secreted by a particular strain can degrade the cellulase affecting the microorganism–enzyme compatibility. On the whole, several process parameters must be optimized: substrate concentration, enzyme to substrate ratio, dosage of the active components (β-glucosidase) in the enzymatic mixture, and yeast concentration.[29]

In this work, a methodology for determining parameters integrating a local identifiability analysis to detect the best identifiable parameter subset (which will be obtained with improved reliability), called model-based identifiable parameter determination (MBIPD), is presented. The MBIPD methodology comprises several tasks: the model selection, the tracking of an adequate IG, the iterative PE and identifiability analysis and the accuracy analysis of the estimated parameters.

This article is organized as follows: in section "Parameter determination and assessment of accuracy and identifiability of estimated parameters," the mathematical models and PE problem formulation are introduced. Moreover, an approach for local identifiability analysis of model parameters using SsS is described in detail. Following a framework for the MBIPD is presented in section "Parameter estimation methodology." The case study is introduced in section "Case study: bio-ethanol from cane bagasse by SSF process" and specific information about the conducted experiments as well as the process model is given. The application of the proposed framework, as well as a result evaluation, is discussed in section "Results." Finally, the results are summarized and conclusions are given.

Parameter Identification Methodology Based on Identifiability Analysis and Assesment of Accuracy

Assume that the process can be described by a system of implicit differential and algebraic equations (DAEs) given by Eq. 1:

$$f(\dot{x}(t), x(t), u(t), \theta, t) = 0 \tag{1}$$

$$y(u, \theta, t) = h(x^*(u, \theta, t)) \tag{2}$$

where f is a set of DAEs with continuous first partial derivatives with respect to its arguments, h is a set of relations between the measured response variables $y(t)$ and the dependent state variables $x(t) \in R^{\mathrm{Nx}}$, $u(t) \in R^{\mathrm{Nu}}$ the time-varying input actions or experiment design variables, $\theta \in R^{\mathrm{Np}}$ the unknown parameters, and $t \in [t_0; t_f] \subseteq R$ the time as independent variable. Initial conditions are given as $x(t_0) = x_0$.

A vector of predicted response variables $y(t) \in R^{\mathrm{Ny}}$ is considered in Eq. 2, whose elements are functions of the solution x^* of problem Eq. 1 (obtained by integration of the DAE system); in most cases, h is simply a "selector" function, selecting those state variables that are in fact measured. Usually not all state variables are measured, and thus, the dimension of the corresponding predicted responses $y(u, \theta, t)$ is also smaller than the dimension of the state variables $x^*(u, \theta, t)$, being the former a subset of the latter $y(u, \theta, t) \subseteq x^*(u, \theta, t)$ and $\mathrm{Ny} \leq \mathrm{Nx}$. Additionally, there exists a discrete set of time instances $t_k \in R^{\mathrm{Nm}}$ for the sampling and collection of the predicted response variables $y(u, \theta, t_k) = [y_1(u, \theta, t_k), \ldots, y_{\mathrm{Ny}}(u, \theta, t_k)]^{\mathrm{T}}$.

The model parameters are determined solving the PE problem in Eq. 3. Parameter values of $\theta_1, \ldots, \theta_{\mathrm{Np}}$ are chosen so as to get the best possible fit to the data using the least squares method[33]:

$$\hat{\theta} = \arg \min_{\theta} \left((Y - Y^m)^T \sum\nolimits_y^{-1} (Y - Y^m) \right) \tag{3}$$

where $\hat{\theta}$ is an unbiased estimator containing the best currently available estimate of the true parameter vector, $Y^m = [y^m(t_1)^{\mathrm{T}}, \ldots, y^m(t_{\mathrm{Nm}})^{\mathrm{T}}]^{\mathrm{T}} \in R^{\mathrm{Ny \cdot Nm}}$ is the experimental data vector, Ny is the number of measured variables and Nm is the number of sampling instances t_k when $y^m(t_k) = [y_1{}^m(t_k), \ldots, y_{\mathrm{Ny}}{}^m(t_k)]^{\mathrm{T}}$ $\forall t_k \in [t_1, \ldots, t_{\mathrm{Nm}}]$ is measured; the collection of the predicted response variables $Y = [y(u, \theta, t_1)^{\mathrm{T}}, \ldots, y(u, \theta, t_{\mathrm{Nm}})^{\mathrm{T}}]^{\mathrm{T}} \in R^{\mathrm{Ny \cdot Nm}}$ are calculated in each sampling instant t_k, with $k = 1, \ldots, \mathrm{Nm}$. Finally, $\Sigma_y \in R^{\mathrm{Ny \cdot Nm} \times \mathrm{Ny \cdot Nm}}$ is the variance-covariance matrix of experimental errors in the measured data.

Local identifiability analysis: SsS algorithm

The local identifiability analysis of model parameters is done here based on the sensitivity matrix $S \in R^{\mathrm{Ny \cdot Nm} \times \mathrm{Np}}$:

$$S(\theta) = \left[\frac{\partial y}{\partial \theta}\bigg|_{t_1}, \frac{\partial y}{\partial \theta}\bigg|_{t_2}, \ldots, \frac{\partial y}{\partial \theta}\bigg|_{t_{Nm}} \right]^T \tag{4}$$

where the sub-matrices $\frac{\partial y}{\partial \theta}\big|_{t_k} \in R^{Ny \times Np}$ denote the sensitivities at each instant t_k evaluated in θ. These sensitivities can be generated either by finite differences or integrating the original model along with sensitivity equations.[34] To account for significant differences in the magnitude of parameters and predicted response variables, a normalization is done, such that the element $s_{ij}{}^n$ of the normalized matrix $S^n(\theta) \in R^{\mathrm{Ny \cdot Nm} \times \mathrm{Np}}$ are calculated by Eq. 5:

$$s_{ij}^n = \frac{\partial y_i}{\partial \theta_j} \cdot \frac{\max(|\theta_j|, \theta_{\mathrm{trsh}})}{\max(|y_j|, y_{\mathrm{trsh}})} \tag{5}$$

where θ_{trsh} and y_{trsh} are user defined thresholds (index trsh), e.g., $\sqrt{\varepsilon_{\mathrm{mach}}}$. All parameters with low or nonexisting sensitivities (columns of S with values equal or near to zero), or linearly dependent parameters are not identifiable. In both cases, S is singular or "almost" singular from a numerical point of view. This situation is undesirable, because it reflects near indeterminacy in the parameter estimates, caused by having more parameters than can be reliably estimated from available measurements.[10]

A sensitivity measure δ_j (see Eq. 6) is introduced to assess the individual parameter weight in a least squares PE problem.[13] It is based on the Euclidean Norm of the jth column of S or S^n, respectively (Eq. 4 or Eq. 5), and measures the mean sensitivity of a predicted response variable to a change in the parameter θ_j. A high value of δ_j means that the parameter θ_j has an important impact on the simulation result, whereas values near zero or zero mean that the simulation result does not depend on the parameter θ_j.

$$\delta_j = \sqrt{\frac{1}{\mathrm{Ny} \cdot \mathrm{Nm}} \sum_{i=1}^{\mathrm{Ny \cdot Nm}} S_{ij}^2} \tag{6}$$

In the following, a procedure for the determination of the identifiable parameter subset is presented, which is called SsS algorithm. All corresponding criteria for the SsS algorithm are adapted from the ill-posed parameter selection approach presented in Burth et al.,[10] and Velez-Reyes and Verghese,[9] which is based on the analysis of the approximated Hessian matrix $H_\theta = S^{\mathrm{T}}S$ of problem Eq. 3, and uses EVD.

In this work, the SsS algorithm is based on the sensitivity matrix S of problem Eq. 3 applying a rank-revealing factorization by SVD.[20] The usage of S and SVD (rather than H_θ and EVD) is favorable from a numerical point of view. The SVD always exists and has shown to give better performance for rank-deficient and ill-conditioned matrices.[35] Moreover, although the computation of a SVD is typically more expensive than an EVD, in the context of larger problems with sparse matrices, the SVD can be performed more efficient.[36,37]

The rank-revealing factorization by SVD yields $S = U\Sigma V^{\mathrm{T}}$, where $U \in R^{\mathrm{Ny \cdot Nm} \times \mathrm{Ny \cdot Nm}}$ is a real or complex unitary matrix, $V^{\mathrm{T}} \in R^{\mathrm{Np} \times \mathrm{Np}}$ the conjugate transpose of V is a real or complex unitary matrix, and $\Sigma \in R^{\mathrm{Ny \cdot Nm} \times \mathrm{Np}}$ is a rectangular diagonal matrix with nonnegative real numbers on the diagonal. The diagonal entries Σ_{jj} are the singular values $\sigma_j \geq 0$ of matrix S such as $\sigma_1 \geq \sigma_2 \ldots \geq \sigma_{\mathrm{Np}}$. A criterion for the nearness to singularity of S is the condition number, $\kappa(S) = \sigma_1/\sigma_{\mathrm{Np}}$. A "very high" $\kappa(S)$ indicates an almost singular as well as ill-conditioned sensitivity matrix. According to Grah,[11] an upper condition number bound ($\kappa^{\max} \cong 1{,}000$) is defined and local parameter identifiability is given, when $\kappa \leq \kappa^{\max}$ holds.

Additionally, the collinearity index, $\gamma(S) = 1/\sigma_{\mathrm{Np}}$, is considered as singularity measurement for the sensitivity matrix S. $\gamma(S)$ reaches infinity if the columns are linearly dependent. In Brun et al.,[13] an empirically found threshold of the collinearity index has been named as $\gamma^{\max} \cong 10$–15.

Thus, if $\gamma(S) > \gamma^{max}$, the corresponding parameter set is considered as poorly identifiable.

In the algorithm for SsS (see Figure 1), the set dimension r of the identifiable parameter subset is obtained, which corresponds to the rank of the sensitivity matrix. The identifiable parameter subset $\tilde{\theta}^{(r)} \cdot \in R^r$ consists in those parameters, which are associated with the highest singular values. All parameters associated with the smallest remaining singular values $\tilde{\theta}^{(Np-r)} \in R^{Np-r}$ have the linear dependency feature. The SsS algorithm depicted in Figure 1 uses six steps described as follow:

1. Compute the SVD of $S(\theta)$ for the current parameter set θ.
2. Calculate singularity measurements: condition number and collinearity index, $\kappa(S)$ and $\gamma(S)$, respectively.
3. Evaluate the singularity measurements and check if the statements $\kappa(S) \le \kappa^{max}$ and $\gamma(S) \le \gamma^{max}$ are fulfilled, if yes, the analyzed parameter vector θ is identifiable and the algorithm finishes, if no, go to the next step.
4. Build a set $\Gamma = \{\sigma_j | \ \kappa_j = \sigma_1/\sigma_j, \wedge, \kappa_j \le \kappa^{max}\}$, such that κ_j is the sub-condition number corresponding to each σ_j available in the diagonal matrix Σ computed in step 1, with $j = 1,\ldots,$ Np.
5. Determine r as the dimension of Γ, such that r is the maximum number of nonzero singular values σ_j of $S \in R^{\mathrm{Ny\cdot Nm\times Np}}$ fulfilling the requirement $\kappa_j \le \kappa^{max}$.
6. Determine a permutation matrix Π by constructing a QR decomposition with column pivoting (QRP) for $S \in R^{\mathrm{Ny\cdot Nm\times Np}}$ such that $S\Pi = QR$, where $Q \in R^{\mathrm{Ny\cdot Nm\times Ny\cdot Nm}}$ is an orthogonal matrix, $R \in R^{\mathrm{Ny\cdot Nm\times Np}}$ is an upper triangular matrix with decreasing order of diagonal elements and $\Pi \in R^{\mathrm{Np\times Np}}$ is a permutation matrix* which orders the columns of S according to linear independency, it means that the first columns of $S\Pi$ are the largest independent set of columns of S.
7. Use Π to reorder the parameter vector θ according to $\tilde{\theta} = \prod^{T}\theta$.
8. Make the partition $\tilde{\theta}=[\tilde{\theta}^{(r)T}, \tilde{\theta}^{(Np-r)T}]^T$ with $\tilde{\theta}^{(r)}$ containing the first r elements of $\tilde{\theta}$, which are the identifiable or "active" parameters (active subset), and $\tilde{\theta}^{(Np-r)}$ containing the last $Np-r$ elements of $\tilde{\theta}$, which are not identifiable or nonactive parameters (nonactive subset).

Assessment of parameter accuracy

A statistical analysis is performed to assess the reliability and adequacy of the parameter estimates. The analysis of the sum of the squared residual in Eq. 3 allows evaluating if the search of the true parameter value θ_j (approximated by its estimation $\hat{\theta}_j$) was successful. Inappropriate initial guesses, parameter correlations or trapping in local minima may cause a termination of the search far from the true minimum θ_j. The accuracy of the estimator $\hat{\theta}$ is reflected by the parameter variance-covariance matrix $\Sigma_\theta \in R^{\mathrm{Np\times Np}}$, Ref. 33:

$$\sum\nolimits_\theta = \begin{pmatrix} \sigma^2_{\theta_1,\theta_1} & \sigma^2_{\theta_1,\theta_2} & \cdots & \sigma^2_{\theta_1,\theta_{Np}} \\ \sigma^2_{\theta_2,\theta_1} & \sigma^2_{\theta_2,\theta_2} & \cdots & \sigma^2_{\theta_2,\theta_{Np}} \\ \vdots & \vdots & & \vdots \\ \sigma^2_{\theta_{Np},\theta_1} & \sigma^2_{\theta_{Np},\theta_2} & \cdots & \sigma^2_{\theta_{Np},\theta_{Np}} \end{pmatrix} \tag{7}$$

a symmetric matrix whose elements $\sigma^2_{\theta_i,\theta_j}$ are the covariance of the ith parameter regarding the jth parameter (with $i, j = 1,\ldots,$ Np) and the diagonal elements $\sigma^2_{\theta_j,\theta_j}$ are the variance of the jth parameter.

According to Cramér-Rao,[33] the lower bound of Σ_θ is calculated by the inverse of the FIM, FIM $\in R^{\mathrm{Np\times Np}}$, which depends on the sensitivity matrix given in Eq. 4 and the variance-covariance matrix of measurements $\Sigma_y \in R^{\mathrm{Ny\cdot Nm\times Ny\cdot Nm}}$:

$$\mathrm{FIM} = S^T \cdot \sum\nolimits_y^{-1} \cdot S \tag{8}$$

Using this approximation: $\sum_\theta = \mathrm{FIM}^{-1}$, Ref. 17, if the sensitivity matrix is normalized by Eq. 5, the elements of Σ_θ then indicate relative rather than absolute uncertainties of the parameters.

Two statistical procedures are considered to assess the model adequacy:

1. Statistical significance of the true parameter value θ_j, which tests if θ_j is zero and could be dropped from the model.
2. Confidence interval (CI), which provides a range of likely values for θ_j for a defined confidence level (α).

The statistical significance of the parameter $\theta_j \in \theta$ with $j = 1,\ldots,$ Np is obtained by a hypothesis-testing procedure. The hypotheses for testing the significance of any individual parameter, say θ_j, are:

$$\begin{aligned} H_0 &: \theta_j = 0 \\ H_1 &: \theta_j \neq 0 \end{aligned} \tag{9}$$

H_0 and H_1 are tested based on a t-distribution with Nm – Np degrees of freedom (DoF).[1,2] The test statistic for this hypothesis for parameter j is defined by Eq. 10:

$$t_0 = \frac{\hat{\theta}_j}{\sqrt{\sigma^2_{\theta_j,\theta_j}}} \tag{10}$$

The null hypothesis (H_0) is rejected if $|t_0| > t_{\alpha/2,\mathrm{Nm-Np}}$, meaning that the parameter θ_j contributes significantly to the model. In contrast, very low t_0-values suggest that the parameter could be statistically dropped from the model. It should be noted that sometimes in multiparameter models, a t_0-value might be low because of high correlation between parameters.[2] Accordingly, the t-test is a partial (or marginal) test affected by parameter interactions (we shall return to this in section "Results").

The second measure for model adequacy is the 100 (1– α) % CI on the individual true parameter θ_j, with $j = 1,\ldots,$ Np, where $1-\alpha$ is called the confidence coefficient or the probability of selecting a sample for which the CI contains the true value of θ_j.[1,38]

$$\hat{\theta}_j - t_{\alpha/2,\mathrm{Nm-Np}}\sqrt{\sigma^2_{\theta_j\theta_j}} = L \le \theta_j \le U = \hat{\theta}_j + t_{\alpha/2,\mathrm{Nm-Np}}\sqrt{\sigma^2_{\theta_j\theta_j}} \tag{11}$$

In Eq. 11, $t_{\alpha/2,\mathrm{Nm-Np}}$ is the critical value of the two-tails Student's t distribution for the given confidence level α and Nm – Np degrees of freedom. The parameter CI ($L \le \theta_j \le U$, where L and U are called the lower- and upper-confidence limits calculated by Eq. 11), describes a range of values within which we can be reasonably sure that the true parameter θ_j actually lies. It is a measure of the uncertainty inherent in the estimate $\hat{\theta}_j$. A narrow CI indicates that the effect size is known precisely. In contrast, wide intervals indicate little knowledge about the true parameter, and that further information is needed. Another interpretation of 95% CIs says that for an infinite number of solutions of the PE problem using the same

*In the QRP algorithm, the permutation Π is determined choosing the maximum column norm of S.

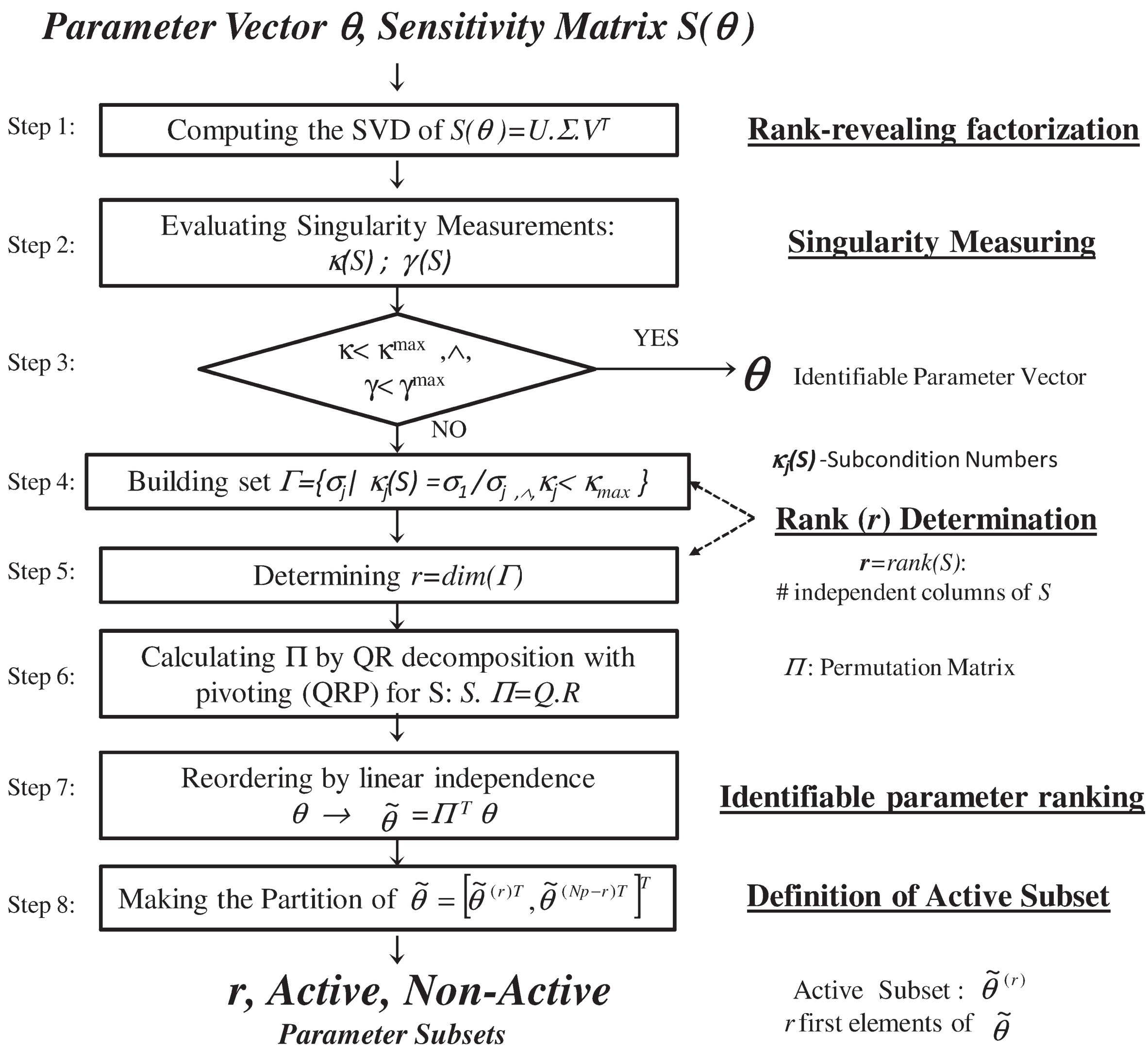

Figure 1. Parameter identifiable subset selection (SsS) algorithm.

experimental conditions and generating the same experimental data but considering random measurement errors (normally and independently distributed with mean zero and constant variance), the true value of the parameter θ_j will be contained within 95% of all observed CIs. More briefly, it can be said that θ_j is between [L, U] "with a 95% confidence."

Parameter estimation methodology

The framework for the MBIPD is depicted in Figure 2. The algorithm starts with model and IG selection, followed by an iterative procedure for PE and SsS. The integration of the algorithm for SsS (Figure 1) permits to find those parameters which are actually identifiable in the system (reducing the dimension of the parameter vector) according to their linear independency. The new reduced problem is then well-conditioned and its solution has a better accuracy than the initial problem including all parameters and their correlations.

Accordingly, when SsS is applied, ill-conditioned parameters are fixed at prior estimates (referred to as "inactive" parameters) and reduced-order problems are considered for the determination of the remaining "active" parameters. The reduced problems involve sensitivities w.r.t. the r < Np remaining or active parameters; we denote these sensitivities by $S^{(r)} \in R^{\mathrm{Ny \cdot Nm} \times r}$, and note that its columns are just a subset of the columns of S (i.e., $S^{(r)} \subseteq S$).

In Figure 2, the methodology is divided in several steps:

1. Select an appropriate process model according to the specific phenomena of the studied system using the experimental data ($Y^m \in R^{\mathrm{Ny \cdot Nm}}$). For doing so, a fitting comparison between several proposed models (f_i, with $i = 1,\ldots, N_{\mathrm{mod}}$) is carried out, where the model with the smallest residual (minimum value of objective function, OF, in Eq. 3) is chosen.
2. Define an IG for the initialization of the MBIPD algorithm either using data from literature, by performing a priori calculations using the model, or by analysis of solutions of different PE problems using several IGs as starting points. The latter can be based, e.g., on a sampling type algorithm to obtain initial guesses within a prescribed parameter range. From the different solutions, the result with the best fitting is selected as initial parameter guess (θ_0).
3. Initialize the algorithm with the iteration counter $k = 1$; the set dimension $r_{k=0}=r_0$ is set to the number of parameters Np, which means that all parameters are considered as active parameters. Parameter vector θ is initialized with θ_0

Table 1. Experimental Conditions for Bio-Ethanol Production from Sugarcane Bagasse in the SSF Process

Experiment		Cellulignin Dry Solid Content (w/w)	Pre-Hydrolysis Time (h)	Enzymatic Load* GC-220 (FPU/g)	Enzymatic Load* β-Glucosidase (UI/g)	Initial Yeast, C_{x0} (g/L)	Experiment Duration (h)
1	E1	30%	12	26	17	6	33
2	E2	20%	8	26	27.3	6	30
3	E3	20%	8	26	–	6	33
4	E4	20%	12	30	–	2	33
5	E5	30%	12	35	–	3	33

*Respect to cellulignin dry mass (g).

and partitioned in active and nonactive vectors (i.e., $\theta = [\theta^{r_0 T}, \theta^{\mathrm{Np}-r_0^T}]^T$. Accordingly, the vector of active parameters $\theta^{(r0)} \in R^{r0}$ contains all Np elements.

4. Solve the PE problem only for the active parameter vector θ^{rk-1} in Eq. 3 for the iteration counter $k = 1$, and obtain the estimated active parameter vector $\hat{\theta}^{(r_{k-1})}$, and the corresponding sensitivity matrix $S^{(r_{k-1})}$ (active sensitivity matrix). Here, the nonactive parameters of θ (i.e., the last $\mathrm{Np} - r_{k-1}$ elements of θ stored in $\theta^{(\mathrm{Np}-r_{k-1})}$) are fixed and not considered in the PE problem. The updated parameter vector $\hat{\theta} \in R^{\mathrm{Np}}$ is then composed of the r_{k-1} active parameters found as solution of the PE problem (stored in $\hat{\theta}^{(r_{k-1})}$) and the $\mathrm{Np} - r_{k\text{-}1}$ fixed parameters (stored in $\theta^{(\mathrm{Np}-rk-1)}$), i.e., $\hat{\theta} = [\hat{\theta}^{(r_{k-1})\mathrm{T}}, \theta^{(\mathrm{Np}-r_{k-1})\mathrm{T}}]^{\mathrm{T}}$. In the same way, the sensitivity matrix $S(\hat{\theta})$ is composed of the sensitivities corresponding to the r_{k-1} active parameters of $\hat{\theta}$ (columns stored in $S^{(r_{k-1})} \in R^{\mathrm{Ny}\cdot\mathrm{Nm} \times r_{k-1}}$) and the sensitivities of the $\mathrm{Np} - r_{k-1}$ nonactive or fixed parameters of $\hat{\theta}$ (columns stored in $S^{(\mathrm{Np} - r_{k-1})} \in R^{\mathrm{Ny}\cdot\mathrm{Nm}\times\mathrm{Np}-r_{k-1}}$), i.e., $S(\hat{\theta}) = [S^{(r_{k-1})}, S^{(Np-r_{k-1})}]$.
5. Select the new active subset of parameter vector $\hat{\theta}^{(r_{k-1})} \in R^{r_{k-1}}$ applying the SsS algorithm displayed in Figure 1 and using $S^{(r_{k-1})}$. The number of independent columns (uncorrelated parameters) of $S^{(r_{k-1})}$ are determined as the rank (r_k) of this matrix. $\tilde{\hat{\theta}}^{(r_{k-1})}$ is the new ordered parameter vector w.r.t. $\hat{\theta}^{(r_{k-1})}$. The first r_k active elements of $\tilde{\hat{\theta}}^{(r_{k-1})}$, which are the new active parameters, are in $\tilde{\hat{\theta}}^{(r_k)}$, and $\tilde{\hat{\theta}}^{(r_{k-1}-r_k)}$ containing the last $r_{k-1} - r_k$ elements of $\tilde{\hat{\theta}}^{(r_{k-1})}$, which are the new nonactive parameters in the current iteration k. If the singularity measurements (in the current iteration k) are smaller than their thresholds (see section "Local identifiability analysis: SsS algorithm"), the rank of the current iteration (r_k) equals (r_{k-1}), the algorithm finishes and $\hat{\theta}$ (defined in step 4) is the identifiable estimated parameter vector.
6. Run parameter statistical analysis for assessing the accuracy of parameters in the active parameter vector $\tilde{\hat{\theta}}^{(r_k)}$ by calculating their standard deviations, their statistical significance and their CIs based on the approximated parameter variance-covariance matrix $\Sigma_\theta = \mathrm{FIM}^{-1}$ (see Eq. 8).
7. Repeat steps 4–6 with $k = k + 1$ until the SsS applied to the reduced PE problem (Figure 1) indicates a well-conditioned problem.

To verify the obtained solution, the algorithm might be restarted using the final parameter values $\hat{\theta}^{(r_k)}$ as new IG.

Minimum Bias Latin Hypercube Design (MBLHD). To find an appropriate IG (step 2 in Figure 2), a sampling type algorithm within a prescribed parameter range according to MBLHD and the IG selection procedure was applied. MBLHD is based on Latin hypercube sampling (LHS), which provides unique values for each point and exhibits better dispersion than other sampling procedures such as the random and grid sampling.[5,39–41] According to McKay et al.,[39] a LHS is constructed by dividing the range for each of the m input variables into N strata of equal marginal probability $1/N$. If each input variable is assumed to be distributed uniformly on the interval $[-1, 1]$, then the strata are all of width $2/N$. A single value is selected randomly from each stratum, producing N sample values for each input variable. The values are randomly matched to create N sets of values for the N simulator runs.

MBLHD follows the same sampling structure of LHS before but additionally it has found to satisfy the minimum bias conditions of symmetry and orthogonality described by Palmer and Tsui,[41] with sample values for each input variable x_{iu} with $i = 1,\ldots, m$ and sample point $u = 1,\ldots, N$ given by Eq. 12:

$$x_{iu} = \frac{2u-N-1}{\sqrt{(N^2-1)}} \tag{12}$$

where N is the number of total sampling elements, which is chosen for each sample value.

Case Study: Bio-Ethanol from Cane Bagasse by SSF Process

In the experimental case study, parameters of a model for bio-ethanol production from sugarcane bagasse in a SSF process were determined.

Experiments

Table 1 shows the five different experiments E1, E2, E3, E4, and E5 considered in this case study. Experiments E1, E2, E3, and E4 were reported by Vasquez[30] and carried out in the Bioprocess Development Laboratory in chemical school of Federal University of Rio de Janeiro. Experimental conditions of E1–E4 were defined in Vasquez[30] by using the analysis of variance method (ANOVA) in factorial designs (e.g., 2^3 and 3^4 designs) and subsequent optimizations with response surface analysis. Those experiment designs were addressed: (a) to reveal the significant factors on the enzymatic hydrolysis (analyzed factors: temperature, pH, enzymatic load, and cellulignin content), and (b) to investigate the responses of the SSF process for the manipulation of the corresponding three main factors (i.e., cellulignin content, pre-hydrolysis time, and initial yeast). Experiment E5 was achieved in the Bioprocesses Laboratory in the

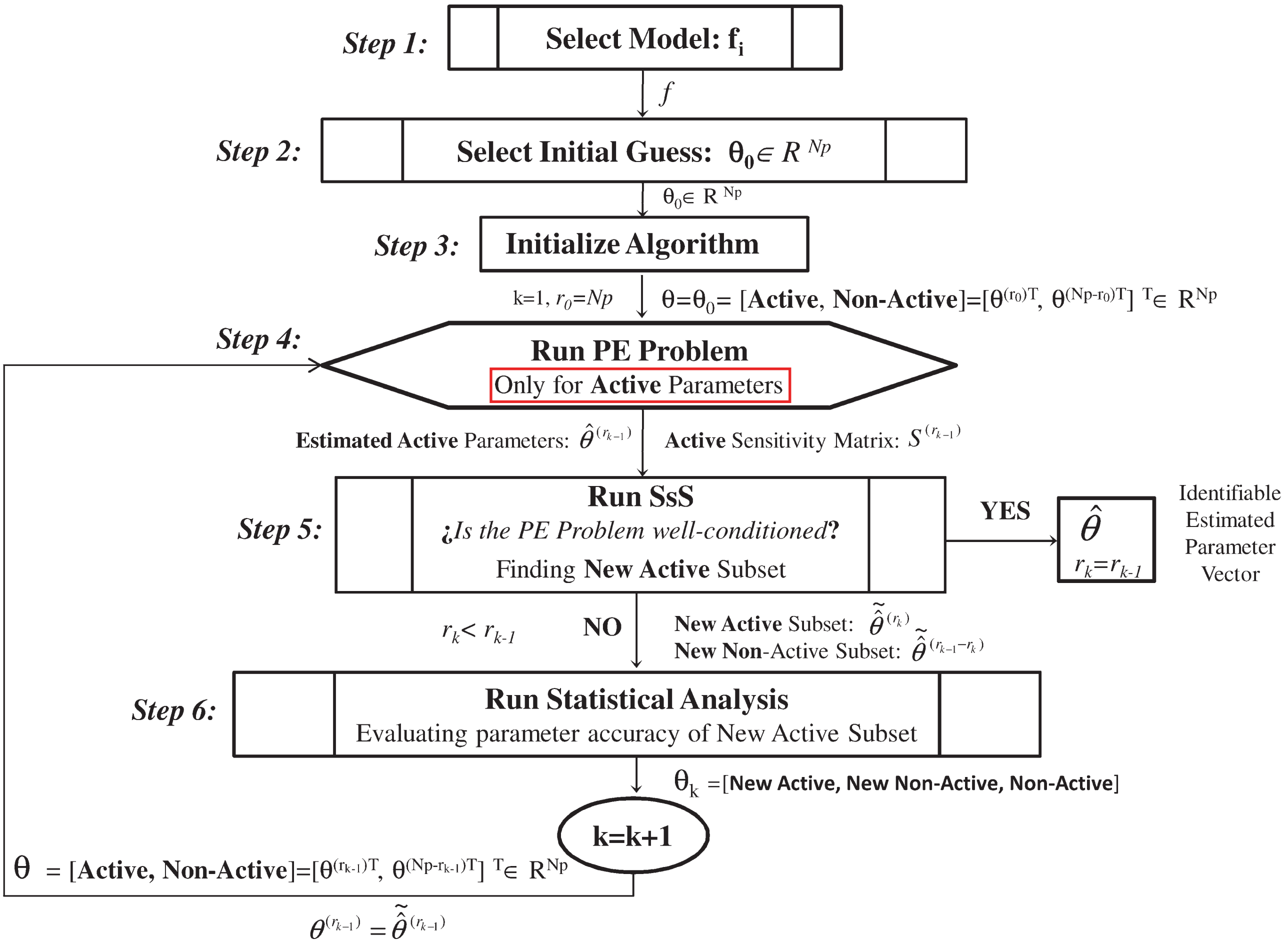

Figure 2. Model-based identifiable parameter determination (MBIPD) algorithm.

Engineering Faculty of University of Antioquia, Colombia; this experiment was designed using the optimal experimental conditions identified by Vasquez [30] for cellulignin content and prehydrolysis time but changing the values of all remaining input factors of the SSF process in Table 1.

Each experiment was repeated twice and a very good repeatability was obtained for all experiments. Experimental data E2 and E3 (E2&E3) were used in the MBIPD algorithm presented above, and experiments E1, E4, and E5 were considered for the validation of the identified model.

The common characteristics of all five experiments are listed below:

a. The lignocellulosic material (cellulignin) was a pretreated residue from sugarcane (i.e., sugarcane bagasse).The cellulignin was obtained by acid hydrolysis of sugarcane bagasse from which the hemicellulosic fraction was removed. This resulting solid residue was pretreated for increasing the accessibility of enzymes to cellulose by a partial removal of the lignin. The pretreatment of lignocellulosic biomass and the measurements of enzymatic load were conducted as described in Vasquez et al.[42]
b. The initial suspension contained dry weight solid composed of the lignocellulosic material considering a content of cellulose of 67% (w/w).
c. Before the SSF process, an enzymatic pre-hydrolysis at 50°C was carried out to allow for the buildup of fermentable glucose.
d. Commercial enzymatic preparations with GC 220–Genencor and β-glucosidase (activities of 104 FPU/mL and 439 IU/mL, respectively) were used; the concentration of protein per mL of GC 220 was 220 mg/mL and 109 mg/mL of β-glucosidase. Enzymatic load in Table 1 makes reference to cellulignin dry mass in gram.
e. After pre-hydrolysis stage, microorganisms were added at 37°C; the microorganism was a commercially available *Saccharomyces cerevisiae* (Fleischmann' yeast).
f. Glucose, cellobiose, and ethanol concentrations were determined by high-performance liquid chromatography (HPLC) using a ShodexSCIOlI ion exchange column for sugars (300 $\times$ 8 mm^2; Shoko Co., Tokyo) at 80°C as stationary phase and degassed Milli-Q (Molsheim, France) water as the mobile phase at a flow rate of 0.6 mL/min.[30] Standard deviations of cellobiose, glucose, and ethanol concentrations in the HPLC were 0.022 g/L, 0.016 g/L, and 0.020 g/L, respectively.

SSF model

The SSF process is described by a generic dynamic model taken from Drissen et al.,[24] which is a compilation of models of different complexity presented in Drissen et al.[23] and Philippidis et al.[25–27] For parameterizing these models, several experimental substrate-enzyme-microorganism systems were used in literature depending on the goal of the study. For instance, the systems of Avicel-Cellubrix†-*Saccharomyces cerevisiae* (Drissen et al.)[24] and waste paper-Econase‡-*Saccharomyces cerevisiae* (Philippidis and Hatzis)[27] were used

†Enzyme complex of cellulase and β-glucosidase of Novozymes Corp., Denmark.

‡Enzyme complex of cellulase and β-glucosidase of Enzyme Development Corp., New York, NY.

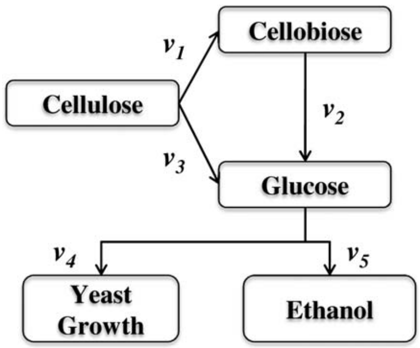

Figure 3. Simplified reaction mechanisms in SSF processes (Drissen et al.).[23]

for parameterizing both process hydrolysis and fermentation, whereas the systems Avicel-Cellubrix[§]-*Saccharomyces cerevisiae* and wheat straw-Cellubrix[¶]-*Saccharomyces cerevisiae* were used only for the hydrolysis stage (Drissen et al.)[23] and the system glucose-*Brettanomyces custersii* system only for the fermentation stage (Philippidis et al.).[25]

The generic dynamic model considers the four most influencing factors for the kinetics of SSF: the cellulosic substrate concentration, the cellulase and β-glucosidase enzyme system, the substrate-enzyme interaction, and the enzyme-yeast interaction. The simplified reaction mechanisms are presented in Figure 3, where cellulose is simultaneously hydrolyzed to cellobiose (v_1 as production rate of cellobiose from cellulose by cellulase) and glucose (v_3 as production rate of glucose from cellobiose by β-glucosidase), cellobiose is then converted to glucose (v_2 as production rate of glucose from cellulose by cellulase), and glucose is catabolized to ethanol, biomass, and carbon dioxide by the fermentative microorganism. Yeast growth and glucose consumption rates are expressed by v_4 (i.e., biomass production rate) and v_5 (i.e., substrate consumption rate), respectively. The hydrolysis of cellulose by cellulase is a reaction that takes place on the surface of the insoluble substrate (heterogeneous catalysis), whereas hydrolysis of cellobiose byβ-glucosidase is carried out in the aqueous phase (homogeneous catalysis).

The dynamic behavior of cellulose (C_c), cellobiose (C_{cb}), glucose (C_G), biomass (C_X), and ethanol (C_{EtOH}) concentration during a batch SSF process are described by the following mass balances in Eqs. (13–17). The change in enzyme concentration (C_E)[23] is presented in Eq. 18.

$$\frac{dC_C}{dt} = -[v_1 + v_3] \tag{13}$$

$$\frac{dC_{Cb}}{dt} = 1.056v_1 - v_2 \tag{14}$$

$$\frac{dC_G}{dt} = 1.053v_2 + 1.111v_3 + v_5 \tag{15}$$

$$\frac{dC_X}{dt} = v_4 \tag{16}$$

$$C_{EtOH} = -0.511.\begin{bmatrix} 1.111(C_c - C_{c0}) + 1.053.(C_{cb} - C_{cb0}) \\ +(C_G - C_{G0}) + C_X - C_{X0} \end{bmatrix} \tag{17}$$

$$\frac{dC_E}{dt} = -K_D C_E \tag{18}$$

The kinetics of the hydrolysis model (v_1, v_2, and v_3 expressed by g L^{-1} h^{-1}) are presented in Eqs. (19–21), in which $k_{max,i}$ with i = 1,…, 3, is the maximum specific rate at full saturation of the substrate with enzyme in the respective reaction. The active amount of enzyme, for reactions with cellulose as substrate (v_1 and v_3), is assumed to be determined by enzyme adsorption onto the cellulose substrate according to the principles of heterogeneous catalysis (Langmuir adsorption constant K_L). Inhibition by glucose to cellulase and β-glucosidase was assumed through $K_{1,G}$ and $K_{2,G}$. For all three reactions, v_1, v_2, and v_3 in Figure 3, the zero-order rate constant is given as a function of temperature (activation energy E_a); in addition, all enzyme activity is assumed to be subject to thermal inactivation (K_D). Moreover, inhibition of cellulase by ethanol is assumed to affect the rates of reaction v_1 and v_3 by inhibition constant $K_{1,EtOH}$.[24] The thermal inactivation constant (K_D) follows an Arrhenius type relationship, $K_D(T) = A_D \cdot e^{-\Delta H/T}$. Reductions in the glucose production originated by changes in nature of the cellulose substrate during the SSF process is considered using a recalcitrance constant Krec.[24]

$$v_1 = \left(k_{max,1} \cdot \frac{C_E}{K_L + C_E}\right).C_c \cdot \left(\frac{K_{1,G}}{K_{1,G} + C_G}\right) \cdot \left(e^{-K_D(T).t}\right) \cdot \left(\frac{e^{\frac{-Ea}{RT}}}{e^{\frac{-Ea}{RT_{ref}}}}\right) \cdot \left(\frac{K_{1,EtOH}}{K_{1,EtOH} + C_{EtOH}}\right) \cdot \left(e^{-Krec.\left(1 - \frac{Cc}{Cc_0}\right)}.\right) \tag{19}$$

$$v_2 = (k_{max,2}.e_g.e_T) \cdot \left(\frac{C_{cb}}{K_m\left(1 + \frac{C_G}{K_{2,G}}\right) + C_{cb}}\right) \cdot \left(e^{-K_D(T).t}\right) \cdot \left(\frac{e^{\frac{-Ea}{RT}}}{e^{\frac{-Ea}{RT_{ref}}}}\right) \tag{20}$$

$$v_3 = \left(k_{max,3} \cdot \frac{C_E}{K_L + C_E}\right).C_c \cdot \left(\frac{K_{1,G}}{K_{1,G} + C_G}\right) \cdot \left(e^{-K_D(T).t}\right) \cdot \left(\frac{e^{\frac{-Ea}{RT}}}{e^{\frac{-Ea}{RT_{ref}}}}\right) \cdot \left(\frac{K_{1,EtOH}}{K_{1,EtOH} + C_{EtOH}}\right) \cdot \left(e^{-Krec.\left(1 - \frac{Cc}{Cc_0}\right)}.\right) \tag{21}$$

For modeling glucose consumption and biomass formation, standard Monod kinetics expanded to include ethanol inhibition on yeast[24] were assumed. Yeast growth rate (v_4) and substrate consumption rate (v_5) are described in Eqs. 22 and 23.

$$v_4 = \frac{\mu_{max}.C_G}{K_G + C_G}.C_x \cdot \left(\frac{K_{iy,EtOH}}{K_{iy,EtOH} + C_{EtOH}}\right) \tag{22}$$

$$v_5 = -\frac{v_4}{Y_{xg}} - C_x.m_s \tag{23}$$

In the above depicted model (Eqs. (13–23)), neither the inhibition by released compounds on cell lysis or secreted by a particular strain on cellulases, nor the inhibition by components in the enzyme preparation which might reduce microbial viability leading to cell lysis[29] were considered. Moreover, the inhibition by some toxic substances arising from pretreatment of the lignocellulose on fermenting microorganisms, as well as on the cellulase activity[29] was also not formulated in the mentioned model.

[§]Enzyme complex of cellulase and β-glucosidase of Novozymes Corp., Denmark.

[¶]Enzyme complex of cellulase and β-glucosidase of Novozymes Corp., Denmark.

Table 2. Initial guesses taken from literature and generated by MBLHD

Parameter		Unit	$IG_{Drissen}$ θ_0	$IG_{Philippidis}$ θ_0	IG_4 θ_0	IG_{22} θ_0
1	$k_{max,1}$	h^{-1}	0.081	0.0827	3.49	21.50
2	$k_{max,2}$	$g\ U^{-1}\ h^{-1}$	0.0108	0.00406	4.42	0.92
3	$k_{max,3}$	h^{-1}	0.058	0.0834	11.5	8.5
4	m_s	h^{-1}	0.02	0	0.884	0.183
5	Y_{xg}	$g\ g^{-1}$	0.11	0.113	0.050	0.750
6	μ_{max}	h^{-1}	0.25	0.19	0.582	2.083
7	K_G	$g\ L^{-1}$	0.0252	0.000037	150	683
8	K_L	$FPUL^{-1}$	18.2	544.89	984	917
9	$K_{1,G}$	$g\ L^{-1}$	6.3	53.16	275	75
10	$K_{1,EtOH}$	$g\ L^{-1}$	95	50.35	15.0	45.0
11	$K_{iy,EtOH}$	$g\ L^{-1}$	50	50	48.3	81.7
12	K_m	$g\ L^{-1}$	1.92	10.56	425	358
13	$K_{2,G}$	$g\ L^{-1}$	0.54	0.62	58.2	491.8
14	$K_{2,EtOH}$*	$g\ L^{-1}$	–	–	225	108

*Parameter added to original model proposed by authors.

Results

All computations were carried out in Matlab 2008®; the solver ode15s was used for integration of the process model, nonlinear regression was performed using nlsqnonlin, an implementation of Levenberg-Marquardt least squares minimization algorithm, which is a hybrid of the Gauss-Newton and the steepest descent method. The parameter set of the DAE system Eqs. (13–23) analyzed in this study is given in Table 2, where the set dimension is Np = 14. An analysis of the concentration measurement showed that the errors for all three analyzed components (cellulose, cellobiose, and glucose) were in the same range with no correlations between measurements. Thus, no specific weighting has been considered and the variance-covariance matrix of experimental errors Σ_y was set to a diagonal matrix using standard deviations reported in section "Experiments."

Measured variables in the experiment conducted by Vasquez[30] were cellobiose (C_{cb}), glucose (C_G), and ethanol (C_{EtOH}) concentrations such that the experimental data vector was $Y^m = [C_{cb}{}^m, C_G{}^m, C_{EtOH}{}^m]^T$ with Ny = 3. E2 and E3 described in section "Case study: bio-ethanol from cane bagasse by SSF process" (Table 1) were considered simultaneously in the PE methodology as discussed in section "Parameter estimation methodology." With 15 sampling points for each measured variable in E2 and 18 points in E3, the total number of sampling points, for the experimental set from now on named E2&E3, is then Nm = 99. Because of the difference between the magnitudes of each parameter involved in this problem, the normalized sensitivity matrix (S^n) was used in the SsS algorithm. The results for the different steps of the MBIPD Algorithm in section "Parameter estimation methodology" and Figure 2 are discussed in the following.

Step 1: selecting model (f)

In the first step, an appropriate model for the SSF was selected. For doing so, differences in the formulation of the SSF models proposed by Drissen et al.,[23,24] and additional changes were analyzed in order to identify the best model structure in terms of the fitting performance (minimum value of the OF). For each structure, several PE problems were solved using data Y^m of all experiments described in Table 1. As result, the model in Eqs. (13–23) was subject to three changes, the first two affecting production rates with cellulose as substrate (v_1 and v_3) and the last one affecting production rates with cellobiose as substrate (v_2):

- Change 1: the factor C_c was removed from Eqs. 19 and 21 and the new lumped parameters $k^*_{max,1}$ and $k^*_{max,3}$ in Eq. 24 with units g L^{-1} h^{-1} are used for production rates v_1 and v_3. This change could be supported by assuming that the hydrolytic system is independent from substrate concentration because all experiments were conducted with higher initial substrate concentrations (C_{c0} taking values of 133.3 and 200 g/L, see Table 1) compared with values in Drissen et al.[23] (C_{c0} = 40 g/L), and therefore, all the time there was an excess amount of cellulignin available for the enzymes.

$$v_i = \left(k^*_{max,i}\cdot\frac{C_E}{K_L + C_E}\right)\cdot\left(\frac{K_{1,G}}{K_{1,G}+C_G}\right)\cdot\left(e^{-K_D(T).t}\right)\cdot\left(\frac{e^{\frac{-Ea}{RT}}}{e^{\frac{-Ea}{RT_{ref}}}}\right)\cdot\left(\frac{K_{1,EtOH}}{K_{1,EtOH}+C_{EtOH}}\right)\cdot\left(e^{-K\mathrm{rec}.\left(1-\frac{C_c}{C_{c0}}\right)}\right)\quad i=1,3 \tag{24}$$

- Change 2: the term making reference to as substrate recalcitrance effect using the parameter Krec, which was considered in Drissen et al.[24] for explaining some reductions in the glucose production according to its experimental data, was removed from Eqs. 19 and 21. The reason for this decision is that a decrease in glucose production by this effect was not expected in the current system due to the fact that the cellulose substrate used in each experiment of Table 1 was highly pretreated, which physically transformed the structure of the sugarcane bagasse (see Vasquez,[30]), making it more susceptible to the enzymatic action, and thus reducing the possibility for this event to take place. Values of Krec close to zero obtained by solving initial PE problems (results here not shown) confirmed this assumption. With this additional change, Eqs. 19 and 21 now become:

$$v_i = \left(k^*_{max,i}\cdot\frac{C_E}{K_L + C_E}\right)\cdot\left(\frac{K_{1,G}}{K_{1,G}+C_G}\right)\cdot\left(e^{-K_D(T).t}\right)\cdot\left(\frac{e^{\frac{-Ea}{RT}}}{e^{\frac{-Ea}{RT_{ref}}}}\right)\cdot\left(\frac{K_{1,EtOH}}{K_{1,EtOH}+C_{EtOH}}\right)\quad i=1,3 \tag{25}$$

- Change 3: Ethanol inhibition proposed by Philippidis et al.[25] is considered using the constant $K_{2,EtOH}$, which affects the rate of the reaction of cellobiose to glucose (v_2) in Eq. 20. This is shown in Eq. 26:

$$v_2 = (k_{max,2}.e_g.e_T)\cdot\left(\frac{C_{cb}}{K_m\left(1+\frac{C_G}{K_{2,G}}\right)+C_{cb}}\right)\cdot\left(e^{-K_D(T).t}\right)\cdot\left(\frac{e^{\frac{-Ea}{RT}}}{e^{\frac{-Ea}{RT_{ref}}}}\right)\cdot\left(\frac{K_{2,EtOH}}{K_{2,EtOH}+C_{EtOH}}\right) \tag{26}$$

The finally selected model including the three changes described above is composed of Eqs. (13–18), 22, 23, 25, and 26. Figure 4 shows the experimental data of E1 and the adjusted predictions of the original model proposed by Drissen et al.[24] (shown in Eqs. (13–23)) and the adjusted predictions of the finally selected model with residual values (OF) of 33.1 and 18.5, respectively.

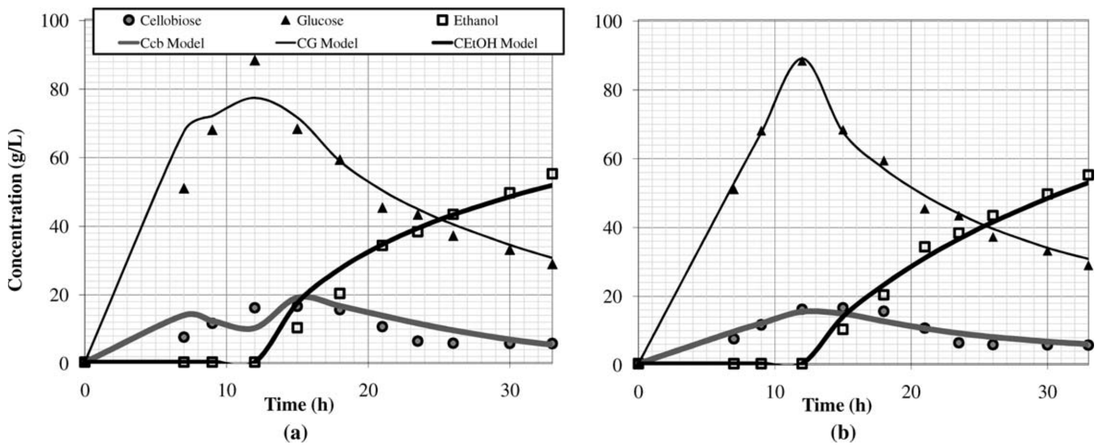

Figure 4. Fitting of experimental data (E1) using: (a) the original model proposed in Drissen et al.,[23] and (b) using the finally selected model after the model selection step.

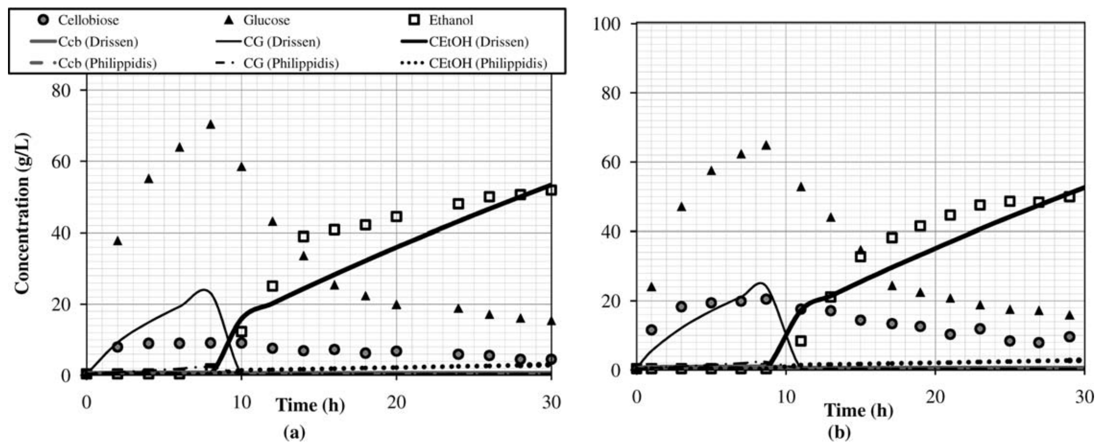

Figure 5. Fitting of experimental data using the model selected in step 1: (a) results for E2 and (b) results for E3. Different initial guesses, $IG_{Drissen}$ see Drissen et al.[24] and $IG_{Philippidis}$ see Philippidis et al.[27], were used for the solution of the PE problems where measured data from E2 and E3 was considered simultaneously.

The parameter values for the thermal inactivation constant $K_D(T) = A_D \cdot e^{-\Delta H/T}$ as well as the activation energy E_a in Eqs. 25 and 26 of the final model were maintained constant according to literature values in Drissen et al.,[24] (ΔH = 1.48 E05 J mol^{-1}, A_D = 3.64 E18 h^{-1}, and E_a = 2.98 E04 J mol^{-1}) taking into account that these parameters had limited impact on the solution according to previous sensitivity analysis. The enzyme concentrations when $t = 0$ (C_{E0}) and values of e_T and e_g depended on the specific enzyme dosage of the specific experiment. C_{E0} was 5200FPU/L for both experiments E2 and E3, whereas values of e_T were 5.35 g/L and 4.58 g/L, and values for e_g were 2,900 U/g and 1,800 U/g for experiment E2 and E3, respectively.

Step 2: selecting initial guess (θ_0)

According to Figure 2, there are different options for defining an initial parameter guess for the initialization of the MBIPD algorithm. In a first attempt, literature values taken from Drissen et al.[24] and Philippidis and Hatzis[27] (referred to as $IG_{Drissen}$ and $IG_{Philippidis}$, respectively) were considered (see Table 2). The solutions of the PE problem (Eq. 3) with the simultaneous consideration of experimental data from E2 and E3 and starting from initial values $IG_{Drissen}$ and $IG_{Philippidis}$ are shown in Figure 5.

It can be seen that extremely low values for C_{cb}, C_G, and C_{EtOH} compared with measured concentrations were calculated after successful termination of the PE algorithm. The worst result is obtained for $IG_{Philippidis}$, with values of the objective function $OF_{IGDrissen}$ = 191.6 and $OF_{IGPhilippidis}$ = 290.5 (see Table 2). The poor fitting might be attributed to remarkable differences between the substrate, enzymatic load, and operating conditions used in the literature experiments compared with experiments E2 and E3. Moreover, differences exist between parameter definitions in models from literature and the selected model because of the differences in the model structure, (e.g., $k_{max,1}$ and $k_{max,3}$ in Table 2 are different from $k^*_{max,1}$ and $k^*_{max,3}$ in Eq. 24); consequently, their true parameter values are far away from that considered in literature.[24,27]

A second strategy for finding an IG was the data collection plan MBLHD (see section "Minimum bias Latin

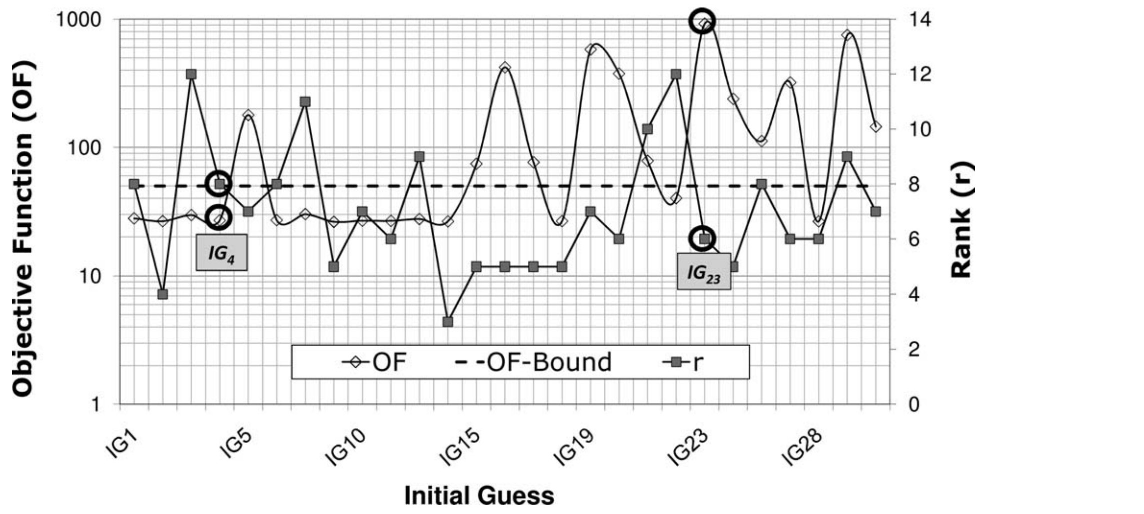

Figure 6. Objective function values and ranks of the sensitivity matrix obtained for the solution of 30 PE problems considering data from E2 and E3 and using different IGs generated by MBLHD.

hypercube design (MBLHD)") for sampling different parameter guesses within a prescribed parameter range was applied. From the parameter range (MBLHD(θ)), a total number of $N_{IG} = 30$ initial guesses were generated and corresponding PE problems were solved. The obtained OF values as well as the corresponding rank (r) of the sensitivity matrix are shown in Figure 6. It should be noted that those points where no convergence could be achieved are not shown in Figure 6 (IG_8, IG_{11}, and IG_{27}). Additionally, the dashed line indicates a user defined upper bound on the OF value, where all IG_i smaller than this bound generated an appropriate fitting of experimental data. It can be seen that the IG_4 was the best IG in terms of the OF value, with OF = 27.0 and $r = 8$. However, in the methodology of identifiable parameter SsS not only the best IG according to the OF value but also according to the highest rank have to be chosen. Thus, IG_{22} with OF = 40.1 and the highest rank $r = 12$ was selected and was used as θ_0 in the next step.

Steps 4 and 5: running iteratively PE and SsS

Figure 7 shows the results of the kth iteration after applying the MBIPD methodology (see Figure 2). A total number of 6 iterations, $k = 1,\ldots,6$, were executed and four parameters (μ_{max}, $k_{max,2}$, K_m, $K_{1,G}$) were finally identified using data from experiments E2 and E3.** In Figure 7, for each iteration, the following results are given: objective function value (OF_k), condition number (κ_k), collinearity index (γ_k), rank of sensitivity matrix (r_k), the current estimated parameter vector ($\hat{\theta}^{(r_{k-1})}$), its variance (σ^2), and its relative standard deviation (σ). The respective last row for the k iteration in Figure 7 contains the result of the SsS, the new ordered parameter vector $\theta = [\theta^{(r_k)^T}, \theta^{(\mathrm{Np}-r_k)^T}]^T$), such that the most identifiable parameter is located at the first position (shady cells contain the elements of identifiable parameter $\theta^{(r_k)}$) and the lowest identifiable parameter is at the last position (elements of $\theta^{(\mathrm{Np}-r_k)}$ include the current nonidentifiable parameters $\hat{\theta}^{(r_{k-1}-r_k)}$ and those nonidentifiable obtained in previous iterations). Figure 8 shows measured data for C_{cb}, C_G, and C_{EtOH} of experiments E2 and E3 and the corresponding predicted concentrations using the parameter vector calculated in iteration $k = 1$ (original parameter vector with 14 estimated elements), in iteration $k = 4$ (7 estimated elements in the reordered parameter vector), and in iteration $k = 6$ (4 estimated elements in the reordered parameter vector).

Figure 9 depicts the ranking of the parameters according to their sensitivity measure δ_j (see Eq. 6). The highest parameter sensitivity measure found in $k = 1$ (i.e., maximum value of δ_j of the columns in the sensitivity matrix $S^{(r0)}$) correspond to K_m and the parameter with lowest sensitivity measure correspond to m_s. It can be seen that the ranking based on sensitivity measures does not give the same results as the ranking according to identifiability (shady cells in Figure 9), e.g., in $k = 1$, the parameter K_G had a medium sensitivity measure but is nonidentifiable. The approach for SsS used in the MBIPD algorithm considers parameters with high sensitivity measures first for the construction of the identifiable parameter subset (using QRP decomposition). It can be seen that for all iterations $k = 1$ to $k = 6$, the parameters K_m, $k_{max,2}$, μ_{max}, and $K_{1,G}$ (which were the most identifiable parameters in the iteration $k = 6$), were in the first positions of the sensitivity ranking, which means that SsS had preferences for selecting the most uncorrelated parameters with high sensitivity. However, some parameters which also affected highly to predicted response variables, e.g., $k^*_{max,3}$ in Figure 9, were not identifiable due to correlations with identifiable parameters (μ_{max}, $k_{max,2}$, K_m, and $K_{1,G}$).

In the first iteration ($k = 1$) with r_0 = Np = 14, after applying the PE and SsS algorithm, the sensitivity matrix rank of the original problem is $r_1 = 12$ (Step 5 in Figure 2), with $OF_1 = 40.1$, $\kappa_1 = 17{,}811$, and $\gamma_1 = 81.3$ (see Figure 7). The singularity measurements confirm that the PE problem is ill-conditioned, despite of the good fitting of the experimental data (see dash-dot line in Figure 8); strong correlations between parameters $\hat{\theta}^{(r_0)}$ indicated by high values of κ_1 and γ_1 were found; the SsS yields the new ordered active parameter vector $\tilde{\hat{\theta}}^{(r_1)} = [K_{1,G}, K_m, k^*_{max,1}, k^*_{max,3}, K_L, Y_{xg}, K_{iy,EtOH}, K_{1,EtOH}, k_{max,2}, K_{2,G}, \mu_{max}, K_{2,EtOH}]^T = \theta^{(r_1)}$ and the nonactive vector $\tilde{\hat{\theta}}^{(r_0-r_1)} = [K_G, m_s]^T = \theta^{(\mathrm{Np}-r_1)}$; thus, the

**It has to be noted that in order to overcome local solutions, the identifiable parameter subset of reduced PE problems was initialized using its corresponding elements from $\theta_0 = IG_{22}$.

k=1, r_0=Np=14			par(1)	par(2)	par(3)	par(4)	par(5)	par(6)	par(7)	par(8)	par(9)	par(10)	par(11)	par(12)	par(13)	par(14)
			$k^*_{max,1}$	$k_{max,2}$	$k^*_{max,3}$	m_s	Yxg	μ_{max}	K_G	K_L	$K_{1,G}$	$K_{1,EtOH}$	$K_{iy,EtOH}$	Km	$K_{2,G}$	$K_{2,EtOH}$
OF_1	40.1	$\hat{\theta}^{(r_0)T}$	11.2	0.097	8.15	2.29E-03	0.337	10.3	883	914	29.7	31.8	9.23	806	13.7	128
κ_1	17811	σ^2	0.28	10.15	0.28	6592.89	1.77	28.86	30.42	0.65	0.17	3.96	17.76	0.12	15.88	31.38
γ_1	81.3	σ	53%	319%	53%	8120%	133%	537%	552%	81%	41%	199%	421%	35%	399%	560%
ρ_1	12	$\theta^{(r_1)T}$	$K_{1,G}$	Km	$k^*_{max,1}$	$k^*_{max,3}$	K_L	Yxg	$K_{iy,EtOH}$	$K_{1,EtOH}$	$k_{max,2}$	$K_{2,G}$	μ_{max}	$K_{2,EtOH}$	K_G	m_s
k=2			par(1)	par(2)	par(3)	par(4)	par(5)	par(6)	par(7)	par(8)	par(9)	par(10)	par(11)	par(12)	$\theta^{(Np-r_1)T}$	
			$K_{1,G}$	Km	$k^*_{max,1}$	$k^*_{max,3}$	K_L	Yxg	$K_{iy,EtOH}$	$K_{1,EtOH}$	$k_{max,2}$	$K_{2,G}$	μ_{max}	$K_{2,EtOH}$		
OF_2	28.1	$\hat{\theta}^{(r_1)T}$	4.51	766	52.8	20.8	909	4.73E-02	28.0	7.48	0.51	1.99	1.74	23.5		
κ_2	22245	σ^2_2	2.17	0.00	2.48	1.76E-03	2.27	206.68	111.32	6.14	0.13	0.76	239.87	7.38		
γ_2	23.7	σ_2	147%	1%	158%	4%	151%	1438%	1055%	248%	36%	87%	1549%	272%		
ρ_2	8	$\theta^{(r_2)T}$	Km	$k^*_{max,3}$	$k_{max,2}$	$K_{2,G}$	$K_{1,G}$	$K_{1,EtOH}$	μ_{max}	K_L	$K_{iy,EtOH}$	$k^*_{max,1}$	Yxg	$K_{2,EtOH}$		
k=3			par(1)	par(2)	par(3)	par(4)	par(5)	par(6)	par(7)	par(8)	$\theta^{(Np-r_2)T}$					
			Km	$k^*_{max,3}$	$k_{max,2}$	$K_{2,G}$	$K_{1,G}$	$K_{1,EtOH}$	μ_{max}	K_L						
OF_3	27.9	$\hat{\theta}^{(r_2)T}$	610	28.3	0.317	2.34	4.12	7.16	1.75	997						
κ_3	1315	σ^2	0.19	2.06	0.37	2.55	0.45	0.32	0.08	1.46						
γ_3	2.6	σ	43%	144%	61%	160%	67%	56%	28%	121%						
ρ_3	7	$\theta^{(r_3)T}$	μ_{max}	$K_{1,EtOH}$	Km	$k_{max,2}$	$K_{1,G}$	K_L	$k^*_{max,3}$	$K_{2,G}$						
k=4			par(1)	par(2)	par(3)	par(4)	par(5)	par(6)	par(7)	$\theta^{(Np-r_3)T}$						
			μ_{max}	$K_{1,EtOH}$	Km	$k_{max,2}$	$K_{1,G}$	K_L	$k^*_{max,3}$							
OF_4	27.9	$\hat{\theta}^{(r_3)T}$	1.74	6.97	593	0.657	3.79	1000	35.7							
κ_4	6593	σ^2	0.06	0.31	0.10	1.79E-03	0.22	11.33	0.05							
γ_4	3.4	σ	25%	56%	31%	4%	47%	337%	23%							
ρ_4	5	$\theta^{(r_4)T}$	$k_{max,2}$	$K_{1,G}$	μ_{max}	$k^*_{max,3}$	Km	$K_{1,EtOH}$	K_L							
k=5			par(1)	par(2)	par(3)	par(4)	par(5)	$\theta^{(Np-r_4)T}$								
			$k_{max,2}$	$K_{1,G}$	μ_{max}	$k^*_{max,3}$	Km									
OF_5	27.9	$\hat{\theta}^{(r_4)T}$	0.657	3.79	1.74	35.7	593									
κ_5	2176	σ^2	3.33E-03	0.06	0.05	0.71	0.03									
γ_5	0.9	σ	6%	24%	23%	84%	16%									
ρ_5	4	$\theta^{(r_5)T}$	μ_{max}	$k_{max,2}$	Km	$K_{1,G}$	$k^*_{max,3}$									
k=6			par(1)	par(2)	par(3)	par(4)	$\theta^{(Np-r_5)T}$									
			μ_{max}	$k_{max,2}$	Km	$K_{1,G}$										
OF_6	28.0	$\hat{\theta}^{(r_5)T}$	1.74	4.56E-03	8.85	3.75										
κ_6	620	σ^2	0.05	7.83E-02	4.23E-07	0.09										
γ_6	0.4	σ	22%	28%	0.07%	30%										
Identifiable Estimated Parameter Vector			par(1)	par(2)	par(3)	par(4)	par(5)	par(6)	par(7)	par(8)	par(9)	par(10)	par(11)	par(12)	par(13)	par(14)
			μ_{max}	$k_{max,2}$	Km	$K_{1,G}$	$k^*_{max,3}$	$K_{1,EtOH}$	K_L	$K_{2,G}$	$K_{iy,EtOH}$	$k^*_{max,1}$	Yxg	$K_{2,EtOH}$	K_G	m_s
$\hat{\theta} = \left[\hat{\theta}^{(r_5)T}, \theta^{(Np-r_5)T}\right]^T$			1.74	4.56E-03	8.8	3.75	35.7	6.97	1000	2.34	28.0	52.8	4.73E-02	23.5	883	2.29E-03

Figure 7. Results of parameter estimation with identifiability analysis.

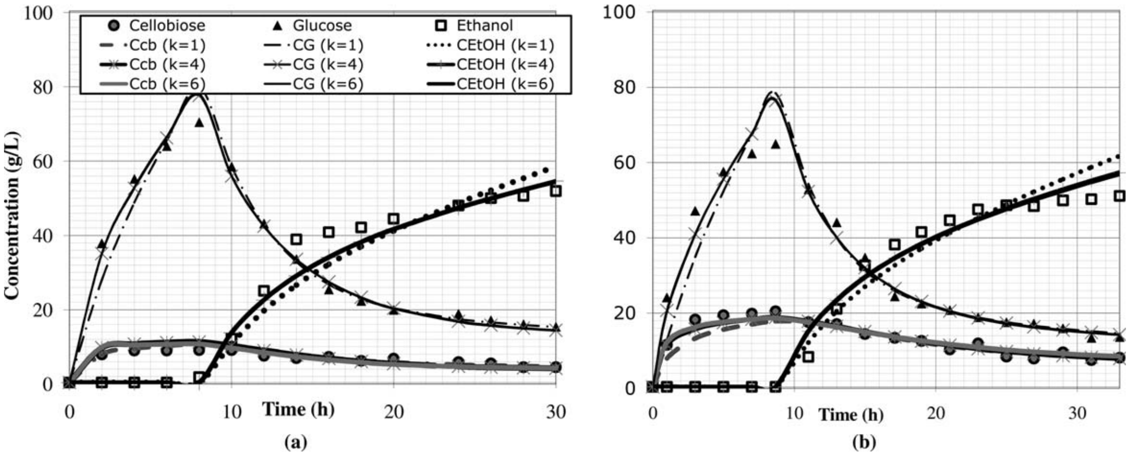

Figure 8. Experimental vs. predicted concentrations using the parameter vector $\hat{\theta}$ calculated in iteration $k = 1, 4, 6$ of the methodology for MBIPD using experimental data of E2 and E3: (a) results for E2 and (b) results for E3.

Sensitivity Ranking	k=1	k=2	k=3	k=4	k=5	k=6	Iteration
	r=12	r=8	r=7	r=5	r=4	r=4	Rank
1	Km	Km	Km	$k_{max,2}$	Km	Km	
2	$k^*_{max,1}$	$k_{max,2}$	$k^*_{max,3}$	$k^*_{max,3}$	$k_{max,2}$	$k_{max,2}$	
3	$k^*_{max,3}$	$k^*_{max,3}$	$k_{max,2}$	Km	$k^*_{max,3}$	$K_{1,G}$	
4	$K_{1,G}$	$K_{1,G}$	$K_{2,G}$	$K_{1,G}$	$K_{1,G}$	μ_{max}	
5	$k_{max,2}$	K_L	$K_{1,G}$	m_{max}	μ_{max}		
6	μ_{max}	$k^*_{max,1}$	$K_{1,EtOH}$	$K_{1,EtOH}$			
7	K_G	$K_{2,G}$	K_L	K_L			
8	$K_{iy,EtOH}$	μ_{max}	μ_{max}				
9	$K_{2,G}$	Yxg					
10	K_L	$K_{1,EtOH}$					
11	$K_{1,EtOH}$	$K_{2,EtOH}$					
12	Yxg	$K_{iy,EtOH}$					
13	$K_{2,EtOH}$						
14	m_s						

Figure 9. Ranking of parameters according to sensitivity during application of MBIPD (most sensitive parameter above in position 1). The identifiable parameter subset in every iteration k is marked by shaded cells.

complete parameter vector for the next iteration $k = 2$ reads $\theta = [\theta^{(r_1)^T}, \theta^{(Np-r_1)^T}]^T$ (see Figure 7) where the shaded cells make reference to the identifiable parameter subset $\tilde{\hat{\theta}}^{(r_k)}$ in iteration k.

In the second iteration ($k = 2$), the 2 nonidentifiable parameters obtained from the iteration $k = 1$ were removed from the PE problem by fixing them to values found at the previous iteration ($K_G = 883$ and $m_s = 2.29$ E -03).

The reduced PE problem was solved and a SsS was performed in the second iteration $\theta^{(r_1)} = \tilde{\hat{\theta}}^{(r_1)} = [k^*_{max,1}, k_{max,2}, k^*_{max,3}, Y_{xg}, \mu_{max}, K_L, K_{1,G}, K_{1,EtOH}, K_{iy,EtOH}, K_m, K_{2,G}, K_{2,EtOH}]^T$, with $r_2 = 8$, $OF_2 = 28.1$, $\kappa_2 = 22{,}245$, and $\gamma_2 = 23.7$. The reduction in γ_2 indicates that this reduced problem is better conditioned than the problem in $k = 1$ but there are still four parameters which are not identifiable, indicated by $r_2 = 8$ and because $\kappa_2 \gg \kappa^{max} = 1{,}000$ and $\gamma_2 > \gamma^{max} = 10$. The identifiable parameter vector reads $\tilde{\hat{\theta}}^{(r_2)} = [K_m, k^*_{max,3}, k_{max,2}, K_{2,G}, K_{1,G}, K_{1,EtOH}, \mu_{max}, K_L]^T$, and the nonactive parameters to be fixed are stored in $\tilde{\hat{\theta}}^{(r_1-r_2)} = [K_{iy,EtOH}, k^*_{max,1}, Y_{xg}, K_{2,EtOH}]^T$. The total vector of nonactive parameters to be held constant at previous estimates in the next iteration $k = 3$, are stored in $\theta^{(Np - r_2)} = [K_{iy,EtOH}, k^*_{max,1}, Y_{xg}, K_{2,EtOH}, K_G, m_s]^T$. The vector of active parameters in $k = 3$ is $\theta^{(r_2)} = \tilde{\hat{\theta}}^{(r_2)}$.

The algorithm in Figure 2 stopped at $k = 6$, with κ and γ being smaller than their defined thresholds ($\kappa_6 < \kappa^{max} = 1{,}000$ and $\gamma_6 < \gamma^{max} = 10$). Consequently, the active parameter subset of this reduced PE problem did not need a further reduction and thus all its estimated parameters $\theta^{(r_5)} = [\mu_{max}, k_{max,2}, K_m, K_{1,G}]^T$ were identifiable with $r_6 = 4$.

Step 6: Running parameter statistical analysis

A statistical analysis of the results obtained in each iteration k was done. The 95% CIs of the estimated parameter vector $\tilde{\hat{\theta}}^{(r_k)}$ were calculated and a significance analysis with a significance level of $\alpha = 0.05$ was performed (see section "Assessment of parameter accuracy"). This is shown in

k=1			$K_{1,G}$	Km	$k^*_{max,1}$	$k^*_{max,3}$	K_L	Yxg	$K_{iy,EtOH}$	$K_{1,EtOH}$	$k_{max,2}$	$K_{2,G}$	μ_{max}	$K_{2,EtOH}$	K_G	m_s
DoF	87	t_0	2.434	2.833	1.879	1.894	1.240	0.752	0.237	0.502	0.314	0.2509	0.186	0.1785		
r_1	12	$\tilde{\hat{\theta}}^{(r_1)T}$	29.7	806	11.2	8.15	914	0.337	9.23	31.8	0.097	13.7	10.3	127.6		
Nm	99	L	5.44	240	-0.65	-0.404	-551	-0.554	-68.1	-94.0	-0.517	-94.6	-99.7	-1294		
α	0.05	U	54.0	1371	23.1	16.7	2380	1.23	86.6	158	0.711	122	120	1549		

k=2			Km	$k^*_{max,3}$	$k_{max,2}$	$K_{2,G}$	$K_{1,G}$	$K_{1,EtOH}$	μ_{max}	K_L	$K_{iy,EtOH}$	$k^*_{max,1}$	Yxg	$K_{2,EtOH}$
DoF	91	t_0	97.20	23.81	2.755	1.148	0.679	0.404	0.065	0.663				
r_2	8	$\tilde{\hat{\theta}}^{(r_2)T}$	766	20.8	0.512	2.0	4.51	7.48	1.74	909				
Nm	99	L	751	19.1	0.143	-1.5	-8.69	-29.4	-52.0	-1814				
α	0.05	U	782	22.6	0.882	5.4	17.7	44.3	55.4	3632				

k=3			μ_{max}	$K_{1,EtOH}$	Km	$k_{max,2}$	$K_{1,G}$	K_L	$k^*_{max,3}$	$K_{2,G}$
DoF	92	t_0	3.600	1.773	2.317	1.653	1.487	0.828	0.696	
r_3	7	$\tilde{\hat{\theta}}^{(r_3)T}$	1.75	7.16	610	0.317	4.12	996.52	28.3	
Nm	99	L	0.78	-0.86	87.1	-0.0639	-1.38	-1393	-52.4	
α	0.05	U	2.71	15.19	1134	0.698	9.62	3386	109.0	

k=4			$k_{max,2}$	$K_{1,G}$	μ_{max}	$k^*_{max,3}$	Km	$K_{1,EtOH}$	K_L
DoF	94	t_0	23.64	2.128	3.973	4.35	3.24		
r_4	5	$\tilde{\hat{\theta}}^{(r_4)T}$	0.657	3.79	1.74	35.7	593		
Nm	99	L	0.602	0.253	0.87	19.4	229		
α	0.05	U	0.713	7.32	2.62	52.0	957		

k=5			μ_{max}	$k_{max,2}$	Km	$K_{1,G}$	$k^*_{max,3}$
DoF	95	t_0	4.394	17.32	6.267	4.232	
r_5	4	$\tilde{\hat{\theta}}^{(r_5)T}$	1.74	0.657	593	3.79	
Nm	99	L	0.96	0.582	405	2.01	
α	0.05	U	2.53	0.733	781	5.56	

k=6			μ_{max}	$k_{max,2}$	Km	$K_{1,G}$
DoF	95	t_0	4.456	3.573	1538	3.302
r_6	4	$\tilde{\hat{\theta}}^{(r_6)T}$	1.74	0.00456	8.85	3.75
Nm	99	L	0.97	0.00203	8.84	1.50
α	0.05	U	2.52	0.00709	8.86	6.01

Figure 10. Statistical analysis: parameter statistical significance (t_j-value) and 95% confidence intervals ($L \leq \theta_j^{(r_k)} \leq U$). Shady cells indicate parameter with statistical significance of 95%.

Figure 10, where DoF is the degree of freedom, r_k is the current parameter vector dimension, Nm the number of experimental data points, t_0 the Student t-value for each jth parameter ($\tilde{\hat{\theta}}_j^{(r_k)} \in \tilde{\hat{\theta}}^{(r_k)}$), L and U are the lower- and upper-confidence limits described by Eq. 11, such that $L \leq \theta_j^{(r_k)} \leq U$.

Shaded cells in Figure 10 indicate significant parameter values in the kth iteration, which were statistically accepted to be different to zero (alternative hypothesis H_1: $\theta_j \neq 0$) with a failure probability of 5%. The low significant parameter quantity at initial iterations demonstrates how parameter correlations influence results of this hypothesis test. For strong correlations, it is not possible to indicate accurately which parameters are not significant. Therefore, those parameters with low t_0-values at the beginning, where correlations are present, should not be interpreted as negligible parameters to the model, but those with higher t_0-values definitely should be considered statistically important.

All components of the last identifiable parameter vector $\hat{\theta}^{(r_5)}$ obtained in iteration $k = 6$, showed a statistical significance of 95%, which also means that their 95% CIs do not include zero ($0.97 \leq \mu_{max} \leq 2.52$, $2.03\ \mathrm{E}\ -3 \leq k_{max,2} \leq 7.09\ \mathrm{E}\ -3$, $8.84 \leq K_m \leq 8.86$, $1.50 \leq K_{1,G} \leq 6.01$ in Figure 10), and therefore, they actually have a strong effect on predicted variables and must remain in the model.

Moreover, to compare the results from the initial problem ($k = 1$, Np = 14) with the last reduced PE problem ($k = 6$, $r_6 = 4$), enormous improvements in the parameter accuracy, validated by reductions in relative standard deviation and narrower CIs, were observed (i.e., an improvement in the maximum relative standard deviation of the most uncertain parameter μ_{max} from 537% to 22% in Figure 7 and corresponding CI from $-99.7 \leq \mu_{max} \leq 120$ to $0.97 \leq \mu_{max} \leq 2.52$ in Figure 10).

Validation of the identified model

The estimated parameter vector $\hat{\theta}$ with 4 identifiable and 10 nonidentifiable parameters shown at the end of Figure 7 fits properly data of two experimental data sets (E2 and E3 in Table 1), which had different experimental conditions:

1. Enzymatic load due to the presence of β-glucosidase,
2. Substrate pretreatment, but also common experimental conditions,
3. Pre-hydrolysis time,
4. Initial cellulose concentration C_{C0}, and
5. Initial yeast concentration C_{X0}.

Accordingly, the fact that 10 parameters of $\hat{\theta}$ had to be fixed might be explained not only by parameter correlations

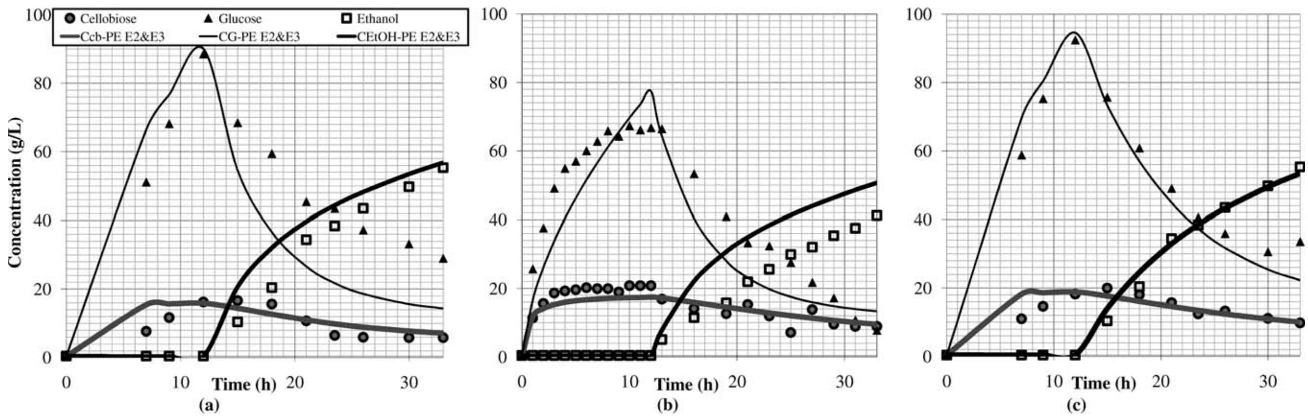

Figure 11. Cross-validation using parameter vectors obtained from PE with E2&E3: experimental vs. predicted concentrations (a) for E1, (b) for E4, and (c) for E5.

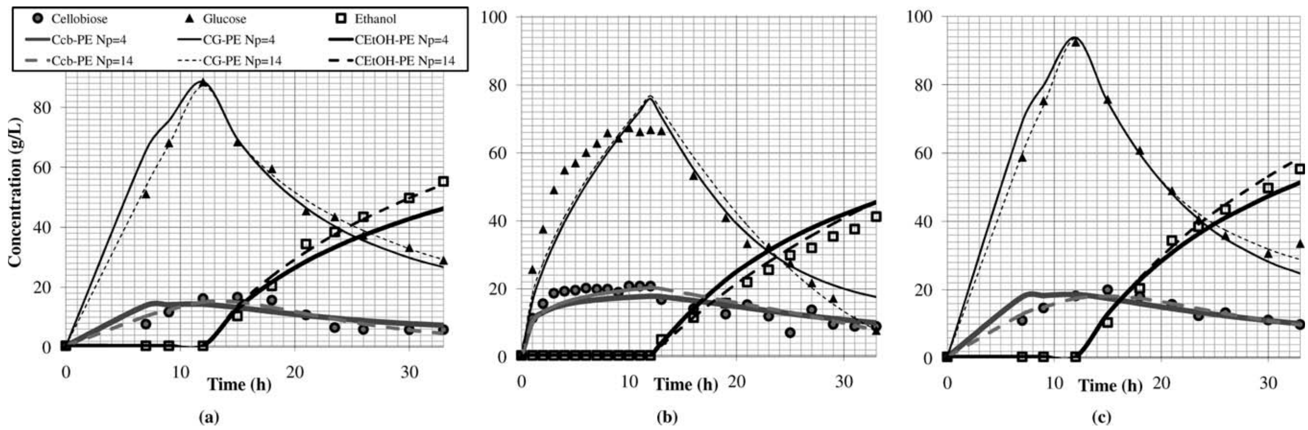

Figure 12. Validation of the identifiable parameter subset using parameter vectors obtained after solving a PE problem with Np = 4 and Np = 14: experimental vs. predicted concentrations (a) for E1, (b) for E4, and (c) for E5.

(nonidentifiability) and weak effects on predicted variables (low sensitivities) but also by the aforementioned common conditions of both experiments. The different and common conditions of E2 and E3 are reflected in the experimental data used for parameter determination. Those shared parameters which are not influenced by the differences are nonidentifiable parameters. In contrast, the 4 identifiable parameters of $\hat{\theta}$ are related to the variability of predicted data for E2 and E3. However, the range of experimental conditions for which the model is valid is certainly limited and structural errors can be partially compensated by adjusting the parameter values. Accordingly, during the validation processes, it is tested whether the identified model is able to predict new experimental conditions and also if the selection of identifiable and nonidentifiable parameter subsets is valid for these conditions.

As a final step of the MBIPD algorithm, a model validation was carried out using experiments E1, E4, and E5 in Table 1; it included:

1. Cross-validation in order to test the validity of model predictions regarding new operating conditions, and
2. Validation of the identifiable parameter subset where the ability to adjust the model predictions to different operating conditions (here E1–E5 in Table 1) is tested by re-estimating the identifiable parameter values only.

In the former analysis, model predictions based on the identifiable estimated parameter vector $\hat{\theta}$ (see Figure 7) were calculated for new experimental conditions which were not considered for parameter determination:

– Cellulose initial concentration C_{C0} (E1 and E5),
– Pre-hydrolysis time (E1, E4, and E5),
– GC-220 and β-glucosidase enzyme load (E1, E4, and E5), and
– Initial yeast concentration C_{X0} (E4 and E5).

These predictions were then compared with experimental data and the quality of fitting was analyzed.

The latter analysis evaluated the fitting of model predictions to different experimental conditions when a PE problem was solved where only the identifiable parameter subset ($\hat{\theta}^{(r_6)}$) was re-estimated. By doing so, the capacity of active parameters to describe the variability in measured data caused by different experimental conditions was assessed.

Cross-Validation. The validity of the model predictions based on the identifiable estimated parameter vector $\hat{\theta}$ (see Figure 7) was tested using experiments E1, E4, and E5 of Table 1. For each experimental condition, SSF process simulations using $\hat{\theta}$ were carried out. The predicted variables are shown as solid lines in Figures 11a–c, respectively. It can be seen that the identified model predicts the behavior under new

Table 3. Estimated parameter vector using E1, E4, and E5 experimental data after solution of different PE problems

		E1			E4			E5		
Parameter		E2 and E3	Np = 4	Np = 14	E2 and E3	Np = 4	Np = 14	E2 and E3	Np = 4	Np = 14
1	$k^*_{\max,1}$	52.84	52.84	3.00	52.84	52.84	58.64	52.84	52.84	4.68
2	$k_{\max,2}$	4.56 E −03	1.37 E −02	2.11 E −01	4.56 E −03	7.81 E −03	3.65 E −01	4.56 E −03	7.03 E −03	2.78 E −01
3	$k^*_{\max,3}$	35.7	35.7	4.4	35.7	35.7	64.2	35.7	35.7	4.9
4	m_s	2.29 E −03	2.29 E −03	7.79 E −04	2.29 E −03	2.29 E −03	4.64 E −01	2.29 E −03	2.29 E −03	2.10 E −05
5	Y_{xg}	0.047	0.047	0.124	0.047	0.047	0.079	0.047	0.047	0.250
6	$\mu_{\max}$	1.74	0.92	2.47	1.74	2.29	2.58	1.74	1.58	7.88
7	K_G	883	883	743	883	883	863	883	883	813
8	K_L	1,000	1,000	167	1,000	1,000	293	1,000	1,000	501
9	$K_{1,G}$	3.75	3.60	220.56	3.75	3.92	1.97	3.75	3.69	109
10	$K_{1,EtOH}$	6.97	6.97	20.4	6.97	6.97	1.54	6.97	6.97	46.3
11	$K_{iy,EtOH}$	28.0	28.0	13.8	28.0	28.0	13.3	28.0	28.0	6.7
12	K_m	8.85	25.80	962	8.85	15.5	823	8.85	14.0	875
13	$K_{2,G}$	2.34	2.34	4.63	2.34	2.34	1.66	2.34	2.34	4.02
14	$K_{2,EtOH}$	23.5	23.5	569.9	23.5	23.5	21.2	23.5	23.5	110
OF		56.4	25.4	9.5	57.1	42.3	35.9	22.6	20.7	9.3

experimental conditions reasonably, with a very good fitting for E5 with residual value of 22.6 and a moderate fitting for E1 and E4 with residual values of 56.4 and 57.1, respectively. During the pre-hydrolysis stage, the model overestimates the cellobiose (C_{cb}—PE E2&E3) and glucose production (C_G—PE E2&E3) for E1 and E5, whereas it underestimates the same concentrations for E4. During the fermentation stage, the model underestimates the glucose production (C_G—PE E2&E3) and overestimates the ethanol production (C_{EtOH}—PE E2&E3) for E1 and E4.

Assessment of the Identifiable Parameter Subset. Glucose (C_G) and ethanol (C_{EtOH}) model predictions for E1 and E4 in Figure 11 demonstrate that the behavior of new experimental conditions could not be predicted properly by the identified model. However, very good predictions are obtained for E5.

A re-estimation of the active or identifiable parameter subset of $\hat{\theta}$ was performed. The identifiable subset (i.e., vector $\theta^{(r_6=4)} = [\mu_{\max}, k_{\max,2}, K_m, K_{1,G}]^T$) was newly estimated for E1, E4, and E5 individually, such that the parameter dimension of the corresponding PE problems was Np = 4.

The fitted model predictions for E1, E4, and E5 are shown as solid lines (Np = 4) in Figures 12a–c, respectively. Comparing the results for E1 and E4 in Figures 12a,b with the results obtained in the cross-validation (see Figures 11a,b), large improvements can be observed, whereas the fitting for E5 in Figure 12c remains almost the same (compare with Figure 11c). These results are reflected by the new residual values for E1, E4, and E5 of 25.4, 42.3, and 20.7, meaning reductions w.r.t. the cross validation residuals of 55%, 26%, and 8%, respectively.

Finally, all parameters were re-estimated (Np = 14) for E1, E4, and E5 individually. Dashed-lines in Figure 12 show the model predictions for Np = 14 which present a better fitting than using those parameters found when Np = 4. Their residual values are 9.5, 35.9, and 9.3 meaning reductions w.r.t. the cross validation residuals of 83%, 37%, and 59% for E1, E4, and E5, respectively. Comparing the results for the PE problems with Np = 14 with the problems with Np = 4, limited improvements regarding the data fitting are visible. These improvements have to be contrasted with a large increase in computation time due to the increased problem dimension. Additionally, these problems are highly ill-conditioned with condition numbers (κ) of 1.08 E 06, 3.10 E 05, and 1.17 E 07 for E1, E4, and E5, respectively (similar to the first iteration step with $k = 1$ of the initial problem in Figure 7).

Table 3 contains the parameter vectors estimated for E1, E4, and E5, the columns "E2 and E3" display the parameter vector $\hat{\theta}$ estimated by the MBIPD algorithm, the columns "Np = 4" display the parameter vector $\hat{\theta}$ when the 4 identifiable parameters (elements of $\hat{\theta}^{(r_6)}$) are re-estimated, and the columns "Np = 14" display the parameter vector $\hat{\theta}$ when all parameters are re-estimated; finally, underlined values make reference to estimated parameters.

Discussion of the results

The model used in this article describes the dynamic behavior of cellulose, glucose, cellobiose, ethanol, and biomass. The cellulose hydrolysis model involved parameters and constants related to the nature and dosage of the enzyme (e_g, e_T, K_D),the enzyme-cellulosic substrate interaction ($k^*_{\max,1}$, $k_{\max,2}$, $k^*_{\max,3}$, K_L, K_m), enzyme-glucose interaction ($K_{1,G}$, $K_{2,G}$) and enzyme-ethanol interaction ($K_{1,EtOH}$, $K_{2,EtOH}$). Moreover, the glucose fermentation model took into account some parameters to describe cell growth ($\mu_{\max}$, K_G), substrate consumption ($K_{iy,EtOH}$), and enzyme-yeast interaction (Y_{xg}, m_s). Thus, differences in the parameter values compared with the values presented by other authors may be explained as by the presence of local minima due to the nonlinearity of the model as by variations in the structural features of the pretreated sugarcane bagasse, variations in the enzymatic mixture (i.e., the addition of pure β-glucosidase to the enzymatic complex GC-220), and variations in the performance of *Saccharomyces cerevisiae* under the specific conditions of the hydrolysis used in this case study.

The accuracy of estimated parameters not only depends on the frequency and accuracy of measurements but also on the selection of measured process variables. Two key variables of the hydrolysis stage were measured (i.e., glucose and cellobiose concentrations), whereas only one key variable of the fermentation stage was measured (i.e., ethanol concentration). According to the experimental fittings, the enzymatic hydrolysis behavior of the experimental data is better described by the model than the fermentative behavior. Consequently, in the fermentation stage, the residuals are bigger in the time region when microorganisms were added; this may be explained by the few experimental information in the fermentation process (i.e., there is no measurement of the biomass concentration which would contribute significantly to the parameter identifiability and might be considered in future works, as additional response variable). Therefore, the fermentation-related parameters were not identifiable and could not be estimated properly. Finally, besides the previously mentioned fact about little experimental information, a more detailed modeling of the fermentative and the hydrolytic process (e.g., including mass transfer phenomena and additional inhibition factors not explored here) should be considered to describe in a more accurate way the real behavior of the process variables.

In this application, the fact that ten parameters were fixed might be explained not only by parameter correlations (nonidentifiability) and weak effects on predicted variables (low sensitivities) but also by the common conditions of both experiments (E2 and E3) displayed in Table 1. In contrast, the four identifiable parameters are related to the variability of predicted data for E2 and E3, to improve the fitting of model predictions to new experimental conditions reflecting the high sensitivity and independence of this selected subset.

The identified model is the result of a model selection step where different literature sources and two different experimental data sets were analyzed. The prediction capacity of this model facing new experimental conditions was assessed and shown in section "Cross-validation." Differences in some of these model output predictions regarding measured variables were evidenced in Figure 11. These fittings were expected taking into account the experimental conditions of new experiments (E1, E4, and E5) were in regions not considered by the experiments used in the parameterization (see E2 and E3 in Table 1). However, proper fittings were also shown in Figure 11, proving that identified model could still predict well new experimental conditions (e.g., fitting of experiment E5).

Limited improvements of data fitting, when PE problems with Np = 14 were solved w.r.t. PE with Np = 4, were obtained in section "Assessment of the identifiable parameter subset." Totally different parameter vectors, a large increase in computation times and ill-conditioned matrices were the features of each PE with Np = 14. According to this evaluation, it is possible to say, the four identifiable parameters had the ability to adapt to new experimental conditions, and therefore, they should be re-estimated for a new experiment only, maintaining the others (unidentifiable parameters) fixed on previous estimate.

Summary and Conclusions

In this contribution, a SSF process model for bio-ethanol production from sugarcane bagasse is identified and critically evaluated. We present a systematic identification methodology, the MBIPD, which is applied to the determination of kinetic parameters. The MBIPD comprises both model and initial guess selection, iterative parameter estimation with local identifiability analysis, and accuracy analysis of the estimated parameters from statistical point of view.

SSF models proposed in literature Drissen et al.[24] and Philippidis and Hatzis[27] are generic formulations for the hydrolysis and fermentation process of any cellulose-enzyme-microorganism system. However, the stated parameter values were identified for a specific cellulosic substrate, enzymatic complex preparation, and microorganisms. Moreover, different phenomena were studied independently in order to estimate individual parameters or parameter subsets. The quality and concentration of the specific cellulosic substrate, enzyme dosage, and mode of substrate-enzyme interactions play a dominant role for the performance of the SSF process. Thus, it was necessary to evaluate the adequacy of literature models and to adapt them (neglecting or adding terms) to the specific characteristics of the precise cellulosic substrate-enzyme-microorganisms system (i.e., pretreated sugarcane bagasse/GC-220 and β-glucosidase/*Saccharomyces cerevisiae* system).

The proposed approach has proved to work efficiently for a strongly over-parameterized model. After successful termination, the number of considered parameters was reduced to a relatively small subset of the original parameter space. Thus, the most influencing parameters for selected operating conditions were identified and the uncertainty of these model parameters was decreased significantly.

The assessment of the identifiability of original parameter vector(with fourteen elements) revealed that a maximum of four model parameters (μ_{max}, $k_{max,2}$, K_m, $K_{1,G}$) are identifiable from the available data (experiments E2 and E3). Uncertainties in the identified parameters expressed by their relative standard deviation was decreased from 537% to 22%, from 319% to 28%, from 35% to 0.07%, from 41% to 30%, for μ_{max}, $k_{max,2}$, K_m, and $K_{1,G}$, respectively. Those huge reductions are also reflected in their smaller CIs compared with the original parameter vector (e.g., CI reduction from $-99.7 \leq \mu_{max} \leq 120$ to $0.97 \leq \mu_{max} \leq 2.52$).

Due to differences between the cellulosic substrate-enzyme-microorganism system used in the literature and here, the use of literature parameters as IG in the MBIPD methodology was not successful. A parameter data collection plan for sampling different parameter guesses within a prescribed parameter range was applied instead (i.e., MBLHD).

It has been shown by a cross-validation that the identified model is able to predict different operating conditions well. The identified model fitted data of two different experimental data sets (E2 and E3) with some common experimental conditions properly. A cross validation using three new experimental data sets (E1, E4, and E5) conducted in different bioprocess laboratories (Federal University of Rio de Janeiro, Brazil, and University of Antioquia, Colombia) was also carried out. Identified model proved to be able to predict the behavior of new experimental conditions with different cellulose initial concentration, pre-hydrolysis time, GC-220, and ß-glucosidase enzyme load and initial yeast concentration.

Identifiable subset enclosed the data variability for new experimental conditions. Moreover, it has been shown that the identifiable parameter subspace should also be used in order to adapt the model to different experimental conditions. The corresponding PE problems can be efficiently solved, as they are of reduced-order and well-conditioned, while giving nearly the same prediction error as if all parameters are considered.

Notation

CI	confidence interval
DAE	differential and algebraic equations
DoF	degree of freedom
E_i	experimental data set i with $i = 1,\ldots,5$
EVD	Eigenvalue decomposition
FIM	Fisher information matrix

IG	initial guess
LHS	Latin hypercube sampling
MBIPD	model-based identifiable parameter determination
MBLHD	minimum bias Latin hypercube design
PE	parameter estimation
QRP	QR decomposition with column pivoting
SSF	simultaneous saccharification and fermentation
SsS	identifiable parameter subset selection
SVD	singular value decomposition
Σ_θ	covariance matrix
A_D	type Arrhenius constant (h^{-1})
C_c	cellulose concentration (g L^{-1})
C_{cb}	cellobiose concentration (g L^{-1})
C_E	enzyme concentration ($FPUL^{-1}$)
C_{EtOH}	ethanol concentration (g L^{-1})
C_G	glucose concentration (g L^{-1})
C_x	yeast concentration (g L^{-1})
e_T	total protein (cellulase and β-glucosidase) concentration per liter reaction volume (g L^{-1})
e_g	β-glucosidase activity per g of protein in the enzyme preparation (U g^{-1})
E_a	activation energy for enzymatic activity (J mol^{-1})
f	set of DAE representing process model
h	set of relations between the measured response variables and the dependent state variables
H_θ	Hessian matrix
K_D	specific rate of cellulose (h^{-1})
K_G	glucose saturation constant for yeast (g L^{-1})
$K_{iy,EtOH}$	inhibition constant of yeast by ethanol (g L^{-1})
K_L	Langmuir adsorption constant (cellulase adsorption saturation constant) ($FPUL^{-1}$)
K_m	Michaelis constant for β-glucosidase for cellobiose (g L^{-1})
$k_{max,1}$	maximum specific rate of cellulose hydrolysis to cellobiose (h^{-1})
$k_{max,2}$	specific rate of cellobiose hydrolysis to glucose (g U^{-1} h^{-1})
$k_{max,3}$	maximum specific rate of cellulose hydrolysis to glucose (h^{-1})
$k^*_{max,1}$	maximum specific rate of cellulose hydrolysis to cellobiose (lumped parameter) (g L^{-1} h^{-1})
$k^*_{max,3}$	maximum specific rate of cellulose hydrolysis to glucose (lumped parameter) (g L^{-1} h^{-1})
Krec	recalcitrance constant
$K_{1,EtOH}$	inhibition constant of cellulase by ethanol (g L^{-1})
$K_{1,G}$	inhibition constants of cellulase by glucose (g L^{-1})
$K_{2,EtOH}$	inhibition constant of β-glucosidase by ethanol (g L^{-1})
$K_{2,G}$	inhibition constant of β-glucosidase by glucose (g L^{-1})
Nm	number of sampling times
N_{mod}	number of available models
Np	number of parameters
Ny	number of measured response variable
Nx	number of dependent state variables
m_s	maintenance requirement for yeast (h^{-1})
OF	objective function value of parameter estimation
Π	permutation matrix of QRP
Q	orthogonal matrix of QRP
r	rank of the sensitivity matrix
R	upper triangular matrix with decreasing diagonal elements of QRP
S	sensitivity matrix
$u(t)$	time-varying input action or experiment design variable
U	real or complex unitary matrix of SVD
V	real or complex unitary matrix of SVD
v_1	production rate of cellobiose from cellulose by cellulase (g L^{-1} h^{-1})
v_2	production rate of glucose from cellobiose by β-glucosidase (g L^{-1} h^{-1})
v_3	production rate of glucose from cellulose by cellulase (g L^{-1} h^{-1})
v_4	production rate of biomass (g L^{-1} h^{-1})
v_5	consumption rate of glucose by yeast (substrate consumption) (g L^{-1} h^{-1}]
$x(t)$	dependent state variable
$y(t)$	measured response variable
$y(u, \theta, t)$	predicted response variable by the model
Y^m	experimental data vector
Y	response variable vector predicted by the model
Y_{xg}	anaerobic yield of cell mass on glucose (yield coefficient of cell mass from glucose) (g g^{-1})
δ_j	sensitivity measure of column j of sensitivity matrix S
ΔH	deactivation enthalpy (J mol^{-1})
γ	collinearity index
γ^{max}	collinearity index threshold
Γ	set of singular values σ_j with $\kappa_j \leq \kappa^{max}$
κ	condition number
κ^{max}	condition number threshold
μ_{max}	maximum growth rate (maximum specific growth rate of the microorganism) (h^{-1})
θ	parameter vector
$\theta^{(Np-r)}$	unidentifiable parameter vector after SsS algorithm
$\hat{\theta}$	unbiased estimate vector
$\theta^{(r)}$	identifiable parameter vector after SsS algorithm
Σ	rectangular diagonal matrix with non-negative real numbers on the diagonal of SVD
σ_j	singular value j

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